

Stabilisation of HIF signalling in the mouse epicardium extends embryonic potential and neonatal heart regeneration

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eLife Assessment

This **valuable** study investigates the role of HIF1a signaling in epicardial activation and neonatal heart regeneration in mice. Using a combination of genetic and pharmacological approaches, the authors demonstrate that stabilization of HIF1a enhances epicardial activation and extends the regenerative capacity of the heart beyond the typical neonatal window following myocardial infarction. The main conclusion is well supported by **solid** data, although some minor concerns regarding experimental interpretation require further clarification to ensure accuracy.

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Abstract

In humans, new-born infants can regenerate their heart during early life. This is modelled in the mouse, where regenerative capacity is maintained for the first week after birth but lost thereafter. Reactivation of this process holds great therapeutic potential; however, the molecular pathways that might be targeted to extend neonatal regeneration remain elusive. Here, we explored a role for hypoxia and HIF signalling on the regulation of epicardial activity in the developing mouse heart and in modulating the response to injury. Hypoxic regions were found in the epicardium from mid-gestation, associating with HIF-1 α and HIF-2 α , and expression of the epicardial master regulator Wilms' tumour 1 (WT1). Epicardial deletion of *Hif1a* reduced WT1 levels, leading to impaired coronary vasculature. Targeting of the HIF degradation enzyme PHD, through pharmacological inhibition with a clinically

approved drug or epicardial-specific deletion, stabilised HIF and promoted WT1 activity *ex vivo*. Finally, a combination of genetic and pharmacological stabilisation of HIF during neonatal heart injury led to prolonged epicardial activation, preservation of myocardium, augmented infarct resolution and preserved function beyond the 7-day regenerative window. These findings suggest modulation of HIF signalling extends epicardial activation to maintain myocardial survival beyond the neonatal regenerative window and may represent a viable strategy for treating ischaemic heart disease.

Introduction

Unlike the adult mammalian heart, the neonatal mouse heart exhibits a remarkable regenerative capacity after myocardial infarction (MI) up to 7 days after birth¹. A similar regenerative capacity was shown in new-born humans². This regenerative potential is linked to the epicardium, a mesothelial layer covering the outer surface of the heart. During development, a subset of epicardial cells undergo epithelial to mesenchyme transition (EMT), generating epicardium-derived cells (EPDCs) that contribute to coronary vessel formation and cardiomyocyte proliferation and compaction³. A number of transcription factors are expressed in the developing epicardium, including Wilms' tumour 1 (WT1), which is instrumental in the regulation of EMT. Loss of WT 1 impairs the ability of EPDCs to migrate into the underlying sub-epicardial region and mutant embryos die between E14.5–E16.5 due to cardiovascular failure⁴. EPDC plasticity gradually decreases during embryogenesis and after birth the epicardium becomes quiescent. Embryonic genes, such as *Wt1*, are downregulated and EMT ceases⁵.

Following injury, the epicardium reactivates, resembling an embryonic-like state. Epicardial cells re-express WT1⁶, proliferate, undergo EMT, and differentiate into fibroblasts to stimulate tissue repair⁷. Thymosin β 4 (T β 4) pre-treatment results in EPDC-mediated neovascularization after injury⁸ and induces limited *de novo* cardiomyocyte formation by differentiation from a progenitor population of epicardial origin⁹.

Hypoxia, or low oxygen levels, has been linked to tissue regeneration¹⁰, and hypoxia-inducible factors (HIFs) play a central role in cellular responses to hypoxia¹¹. HIFs are heterodimeric transcription factors, consisting of α and β subunits¹², with the latter being constitutively expressed, whereas expression of the α subunit is oxygen-dependent (reviewed in¹³). Under well-oxygenated conditions (i.e. normoxia), prolyl hydroxylase domain proteins (PHD1-3) hydroxylate specific conserved proline residues on the α subunit, tagging it for proteasomal degradation. When oxygen levels drop, PHD activity is inhibited, enabling the α subunit to translocate to the nucleus and bind to hypoxia response elements (HRE) regulating transcription of a plethora of target genes¹⁴. In human, mouse and rat there are three isoforms of the α subunit, but the best characterized are HIF-1 α and HIF-2 α activating distinct, although partially overlapping, sets of genes¹⁵. Hypoxia and HIF activation are evident throughout the developing heart¹² and disruption of oxygen sensing pathways during development results in a wide range of cardiac abnormalities¹⁶. Notably, WT1 expression is upregulated in the heart and kidneys of rats exposed to hypoxia and an HRE binding site for HIF-1 α has been identified within the *Wt1* promoter¹⁷, suggesting a link between hypoxia, HIF signaling and epicardial activation. In this context, the role of epicardial HIF signaling in heart development and regeneration remains unclear. Studies have shown that stabilizing HIF-1 α inhibits EPDC migration in avian embryos¹⁸ but promotes their differentiation into vascular smooth muscle cells (VSMCs)¹⁹. Endothelium-specific depletion of *Egln1* and *Egln3* (encoding PHD2 and PHD3, respectively) promotes cardiomyocyte proliferation and prevents left ventricular failure in a model of MI²⁰, and exposure to hypoxia following MI induces a regenerative response in adult mice by metabolic reprogramming of cardiomyocytes leading to cell cycle re-entry²¹.

The epicardium and subepicardial space are identified as hypoxic niches, housing a unique progenitor cell population^{22,23}. Given that hypoxia has been implicated in regenerative responses post-MI, and epicardial WT1 expression is HIF dependent, we hypothesised that HIF signalling contributes to epicardial cell activation during heart development and following injury and as such might represent a therapeutic target for extending the regenerative window²⁴. Here, we demonstrate that the developing epicardium is hypoxic and epicardial-specific deletion of *Hif1a* disrupts epicardial EMT and heart vascularisation. In mice, we observe a gradual and concurrent decrease in WT1 expression and HIF-mediated signalling over the first week of life. Stabilisation of HIF signalling was sufficient to maintain activation of the epicardium beyond postnatal day (P) 7, improve cardiac remodelling and preserve function post-MI. These findings provide novel insight into the molecular mechanism regulating epicardial activation in the mouse heart and suggest that modulation of HIF signalling in the dormant epicardium may enhance cardiac repair and regeneration, representing a potential therapeutic strategy for the treatment of ischaemic heart disease and heart failure.

Results

The epicardium is hypoxic during development

To study the role of low oxygen tension in the developing heart, we initially investigated the physiological levels of hypoxia *in situ* from mid- to late gestation stages, when the epicardium forms and epicardial EMT takes place. Following *in utero* treatment with the marker Pimonidazole-HCl (Hypoxypromide-1, HP1)²⁵, we probed embryonic hearts at various developmental time points, ranging from E12.5 to E18.5 using an anti-pimonidazole fluorescence-conjugated monoclonal antibody (HP1), alongside immunostaining for the spatiotemporal expression of HIF-1 α and HIF-2 α .

At E12.5, characterized by a well-established epicardial layer amidst highly trabeculated myocardium, we observed extensive regions of hypoxia, mostly in areas of dense myocardium, such as the compact wall of the atrioventricular (AV) groove (**Figure 1a**). Notably, HP1 staining co-localized with the epicardial marker WT1 (**Figure 1a**), and both HIF-1 α (**Figure 1-figure supplement 1a**) and HIF-2 α at the apex of the heart. Additionally, strong HIF-1 α signals were noted at the base of the heart (**Figure 1a**), while HIF-2 α expression extended into the myocardial compartment (**Figure 1-figure supplement 2a**).

By E14.5, extensive HP1 staining persisted in areas of thicker compact myocardium, such as the interventricular septum (IVS) (**Figure 1b**). Co-localization with WT1 and HIF-1 α was observed predominantly in the epicardium, notably in the apex (**Figure 1-figure supplement 1b**), and base regions. While HIF-2 α co-localization with HP1 was present, it was primarily within the myocardium (**Figure 1b**, **Supplementary Figure 2b**).

Analysis of hearts at E16.5 and E18.5 revealed a progressive decrease in HP1 signal, becoming largely confined to WT1-expressing epicardial cells at later stages (**Figure 1c, d**). Similarly, HIF-1 α localization was restricted in discrete regions of the outermost layer of the heart (**Figure 1-figure supplement 1c, d**), whereas HIF-2 α expression transitioned from the epicardium (notably at the apex of the heart) to some sporadic expression in underlying cardiomyocytes by E18.5. (**Figure 1-figure supplement 2c-d**, **Figure 1c-d**). Collectively, these data suggest that WT1⁺ cells making up the epicardial layer of the developing heart become hypoxic from mid-gestation onwards, preferentially expressing HIF-1 α whereas HIF-2 α expression predominantly occurs within the forming myocardium.

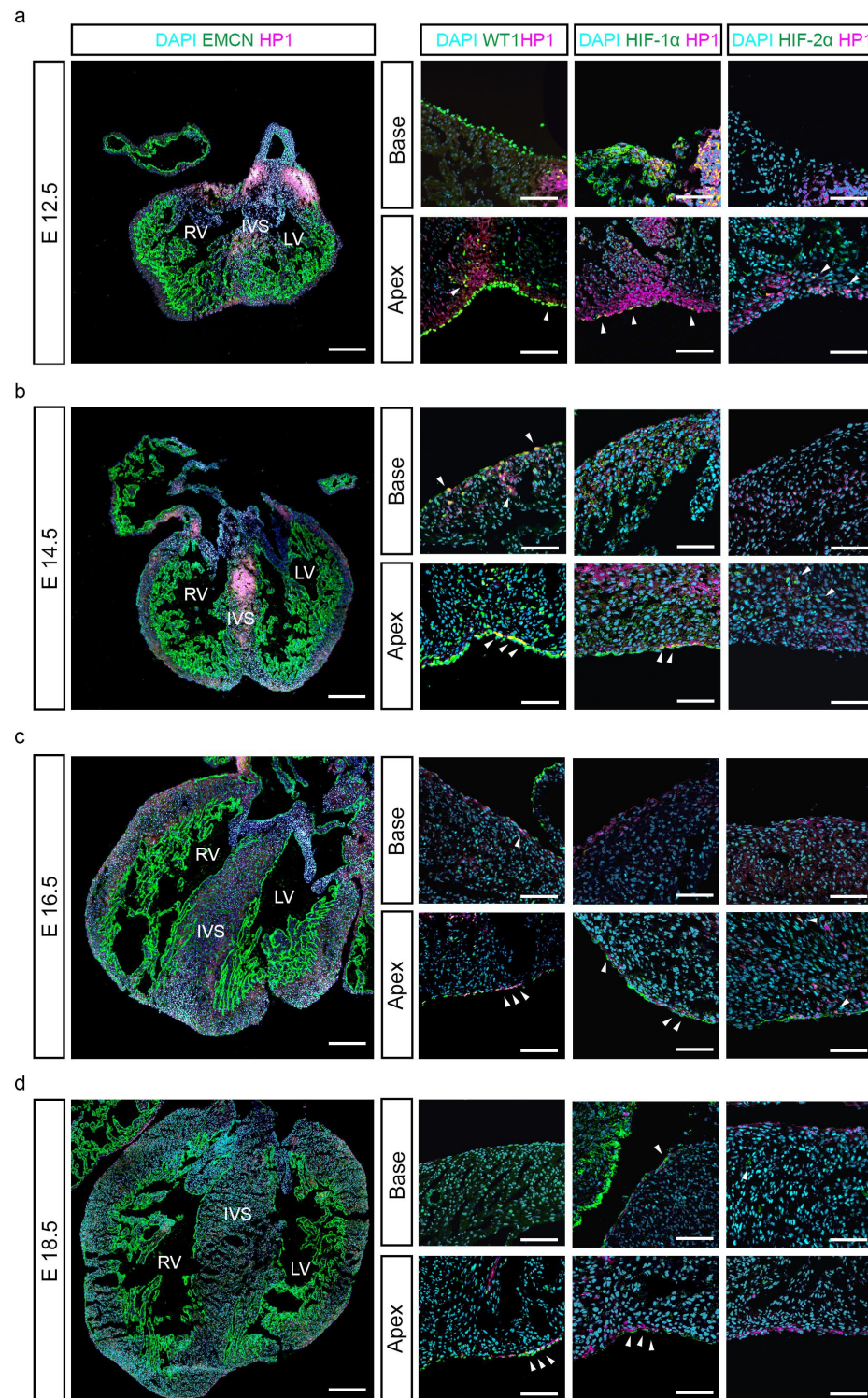


Figure 1.

The epicardium is hypoxic at later stages of heart development.

Representative images of immunostaining for DAPI (cyan), HP1 (magenta) and EMCN, WT1, HIF-1 α and HIF-2 α (green) on serial cryosections of foetal hearts at E12.5 (a), E14.5 (b), E16.5 (c) and E18.5 (d). Arrowheads indicate overlap between HP1 and WT1, HIF-1 α or HIF-2 α . $n = 6$ hearts per stage. IVS= interventricular septum, RV= right ventricle, LV= left ventricle. Whole heart scale bars, 200 μ m; high magnification scale bars, 100 μ m.

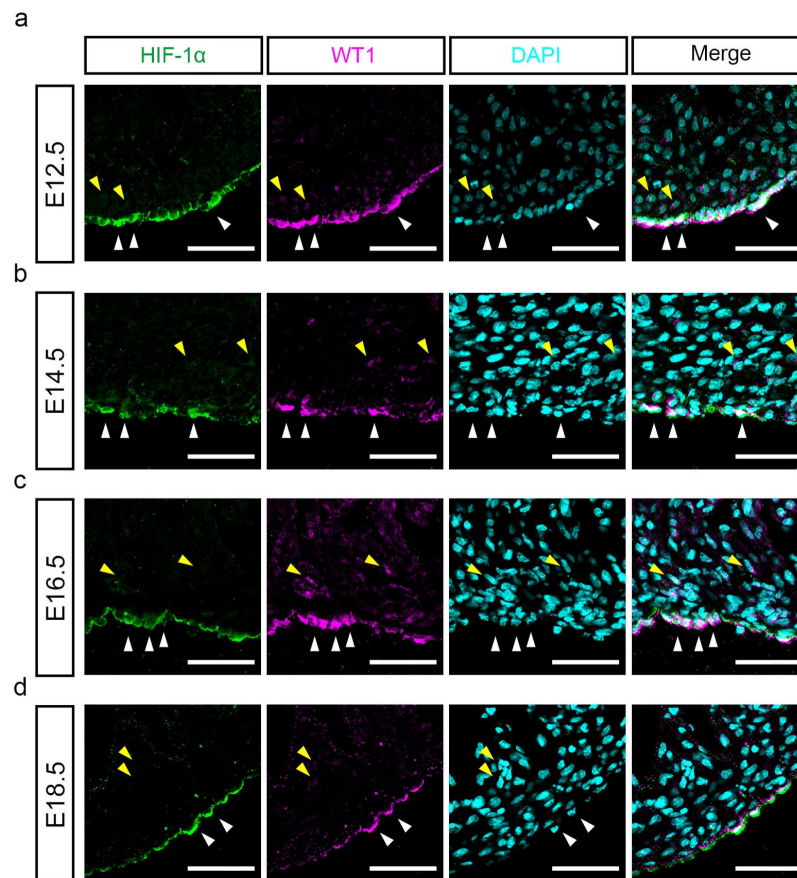


Figure 1-figure supplement 1.

HIF-1α is expressed in the epicardium at different stages of heart development.

Representative images of immunostaining for DAPI (cyan), HIF-1α (green) and WT1 (magenta) on apical regions of foetal hearts at E12.5 (a), E14.5 (b), E16.5 (c) and E18.5 (d). White arrowheads indicate HIF-1α expression in epicardial WT1-expressing cells. Yellow arrowheads indicate WT1-expressing cells lacking HIF-1α expression in the myocardial compartment. $n = 3$ hearts per stage. Scale bar, 50μm.

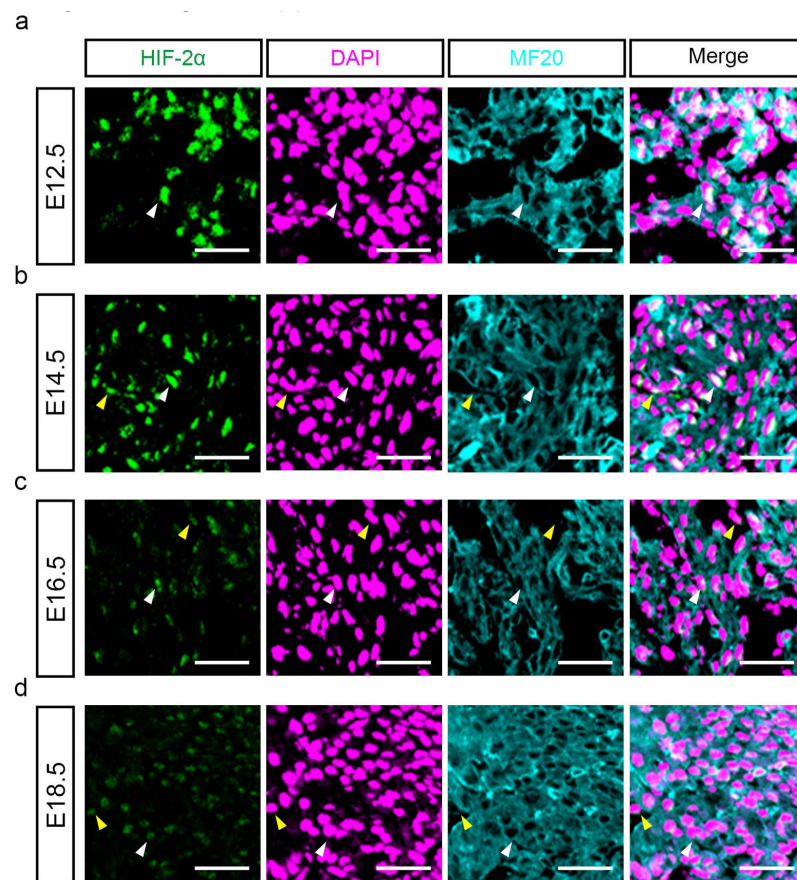


Figure 1-figure supplement 2.

HIF-2α is expressed in the myocardium at different stages of heart development.

Representative images of immunostaining for DAPI (magenta), HIF-2α (green) and MF20 (cyan) on cryosections of foetal hearts at E12.5 (a), E14.5 (b), E16.5 (c) and E18.5 (d). White arrowheads indicate HIF-2α expression in cardiomyocytes. Yellow arrowheads indicate HIF-2α expression in non-cardiomyocytes cells. $n = 3$ hearts per stage. Scale bar, 50μm.

HIF-1 α epicardial deletion reduces the number of Wt1⁺ cells and alters coronary vessel development in E16.5 embryos

Given that the epicardium is hypoxic during mid-late gestation, and we observed expression of HIF-1 α by Wt1⁺ cells within the epicardium, we next determined the functional implications of HIF-1 α on epicardial development. To specifically target *Hif1a* expression in the epicardium, we crossed tamoxifen-inducible epicardial-specific *CreERT2* driver, *Wt1*^{CreERT2/+} mice crossed with mice in which exon 2 of the *Hif1a* gene is flanked by LoxP sites (*Hif1a*^{fl/fl}). Importantly, there is no known phenotype attributed to hemizygoty for *Wt1*²⁷ as might be associated with the *Wt1*^{CreERT2/+} mice²⁶ and as such any interpretation of downstream effect is attributed to epicardial-loss of Hif1 α . Pregnant *Wt1*^{CreERT2/+}; *Hif1a*^{fl/fl} females were injected with tamoxifen at E9.5 and E10.5, to target epicardial development, and embryos were collected at E16.5. Notably, deletion of *Hif1a* did not affect HIF-2 α expression in Wt1⁺ cells (**Figure 2-figure supplement 1a**). Immunostaining for WT1 and the epicardial marker podoplanin (PDPN) revealed a marked reduction in the number of Wt1⁺ cells in *Wt1*^{CreERT2/+}; *Hif1a*^{fl/fl} mutant (KO) hearts as compared to littermate controls (CTR) (**Figure 2a-b**), affecting both the epicardial layer (**Figure 2c**; CTR: 0.72 $\pm$ 0.11; KO: 0.32 $\pm$ 0.010; $p=0.0268$) and the myocardium (**Figure 2d**; CTR: 3.96 $\pm$ 1.01; KO: 0.98 $\pm$ 0.25; $p=0.0461$). Reduction in the number of Wt1⁺ cells in the myocardial compartment suggested an impairment in epicardial EMT. Hearts were also probed for the endothelial marker endomucin (EMCN), which also marks the endocardium and the outline of myocardial trabeculae. No significant alterations were observed in myocardial compaction (**Figure 2e**), myocardial trabeculation as assessed by fractal analysis (**Figure 2f**), or the proliferation rate of Wt1⁺ cells between mutants and controls (**Figure 2-figure supplement 1c-d**).

Given the essential role of the epicardium in coronary vessel development²⁸ we assessed coronary vessel formation using whole-mount immunofluorescence staining for EMCN associated with AngioTool™ software quantification²⁹. Mutant hearts exhibited significantly impaired coronary vasculature, as evidenced by decreased total vessel length (microns) (**Figure 2h**, CTR: 129229 $\pm$ 5509; KO: 81073 $\pm$ 10915; $p=0.0170$), number of junctions (**Figure 2i**, Mean $\pm$ SEM; CTR: 2505 $\pm$ 63.74; KO: 1457 $\pm$ 282.1; $p=0.0223$) and vessel end points (**Figure 2j**, Mean $\pm$ SEM; CTR: 3422 $\pm$ 249.5; KO: 2429 $\pm$ 218.6; $p=0.0402$).

During development, EPDCs undergo EMT and migrate into the sub-epicardial space to support coronary vessel formation. Thus, we sought to investigate whether the vascular impairment observed in the *Wt1*^{CreERT2/+}; *Hif1a*^{fl/fl} mutant embryos was due to compromised epicardial EMT. Epicardial explants from *Wt1*^{CreERT2/+}; *Hif1a*^{fl/fl} embryos treated with tamoxifen showed a reduced HIF-1 α expression (**Figure 2-figure supplement 2b**; CTR: 0.99 $\pm$ 0.16; KO: 0.46 $\pm$ 0.12; $p=0.0397$) without affecting HIF-2 α levels (**Figure 2-figure supplement 2c**; CTR: 2505 $\pm$ 63.74; KO: 1457 $\pm$ 282.1; $p=0.223$). Tamoxifen-treated epicardial explants from *Wt1*^{CreERT2/+}; *Hif1a*^{fl/fl} hearts revealed an increased epithelial and reduced mesenchymal phenotype. This was evident by reduced stress fibre formation (**Figure 2k**), reduced alpha-smooth muscle actin (α -SMA) staining (**Figure 2l**), and altered sub-cellular localisation of the EMT marker and structural component of cell-cell adhesion complexes zonula occludens-1 (ZO-1) (**Figure 2m**) compared to controls. Specifically, mutant explants showed a predominant membrane α -SMA staining, demarcating their epithelial cell shape, whereas control cells displayed a more intense cytoplasmic labelling of the filamentous actin cytoskeleton (**Figure 2l**). Similarly, in control cells the localisation ZO-1 protein was predominantly cytoplasmic and/or nuclear, whereas in mutant cells ZO-1 staining was membranous, indicating a lesser migratory and more epithelial-like phenotype (**Figure 2m**). Consistent with our *in vivo* findings, explants derived from *Wt1*^{CreERT2/+}; *Hif1a*^{fl/fl} embryos exhibited decreased WT1 expression, indicating impaired epicardial EMT (**Figure 2n-o**; CTR: 725.7 $\pm$ 86.93; KO: 490.9 $\pm$ 37.65; $p=0.0479$).

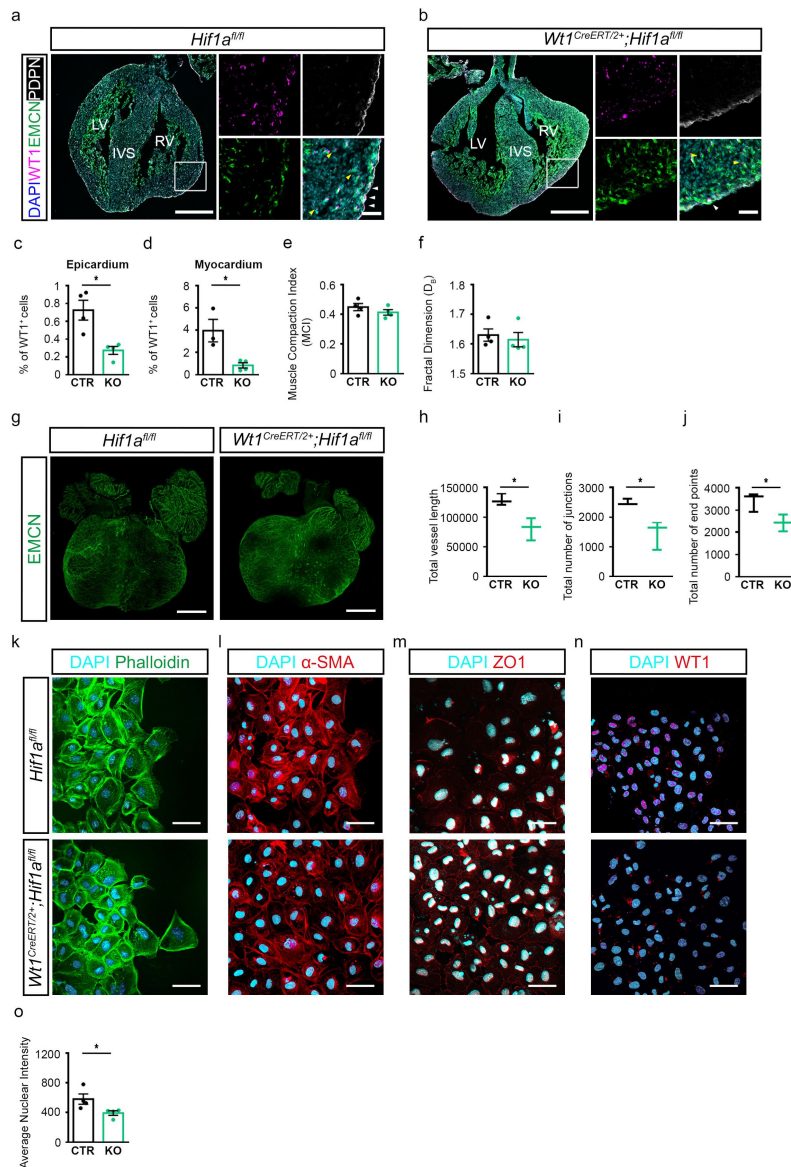


Figure 2.

Epicardial loss of *Hif1α* leads to reduced WT1 expression and impaired EMT.

(a-b) Representative images of immunostaining for EMCN (green), WT1 (magenta), PDPN (white) and DAPI (blue), on coronal sections of hearts from (a) *Hif1α^{fl/fl}* (CTR) or (b) *Wt1^{CreERT2/+};Hif1α^{fl/fl}* (KO) embryos injected with tamoxifen at E9.5 and E10.5 and harvested at E16.5. Images on the right of each panel represent high magnifications of boxed regions. White arrowheads indicate WT1-expressing cells in the epicardium. Yellow arrowheads indicate WT1-expressing cells in the myocardial compartment. Quantification of WT1+ cells in (c) the epicardium (CTR, *n* = 4; KO, *n* = 4) or (d) the myocardium (CTR, *n* = 4; KO, *n* = 4). (e) Muscle compaction index and (f) fractal dimension analysis to assess the complexity of myocardial trabeculae. (CTR, *n* = 4; KO, *n* = 4). (g) Whole-mount immunostaining for EMCN (green) to visualize the coronary vasculature of CTR and KO embryos at E16.5. (h-j) Quantification of coronary vasculature as (h) total vessel length, (i) number of junctions and (j) end points carried out using AngioTool™. *n* = 3 hearts/group. Representative images of immunostaining for (k) Phalloidin (green), (l) a smooth muscle actin (α-SMA, red), (m) zonula occludens-1 (ZO-1, red), (n) WT1 (red) and DAPI nuclear stain (cyan) on epicardial explants derived from *Hif1α^{fl/fl}* (CTR) and *Wt1^{CreERT2/+};Hif1α^{fl/fl}* (KO) hearts harvested at E11.5. (o) Quantification of nuclear intensity of WT1 signal on epicardial explants (CTR, *n* = 4; KO, *n* = 4). IVS= interventricular septum, LV= left ventricle, RV= right ventricle. Whole heart scale bars, 500μm; high magnification scale bars, 50μm. Data presented as median, inter-quartile range (IQR) and upper and lower limits. Error bars represent mean ± s.e.m. *n* numbers refer to individual mice. Two-tailed, unpaired Student t-tests were used for statistical analysis. **p* < 0.05.

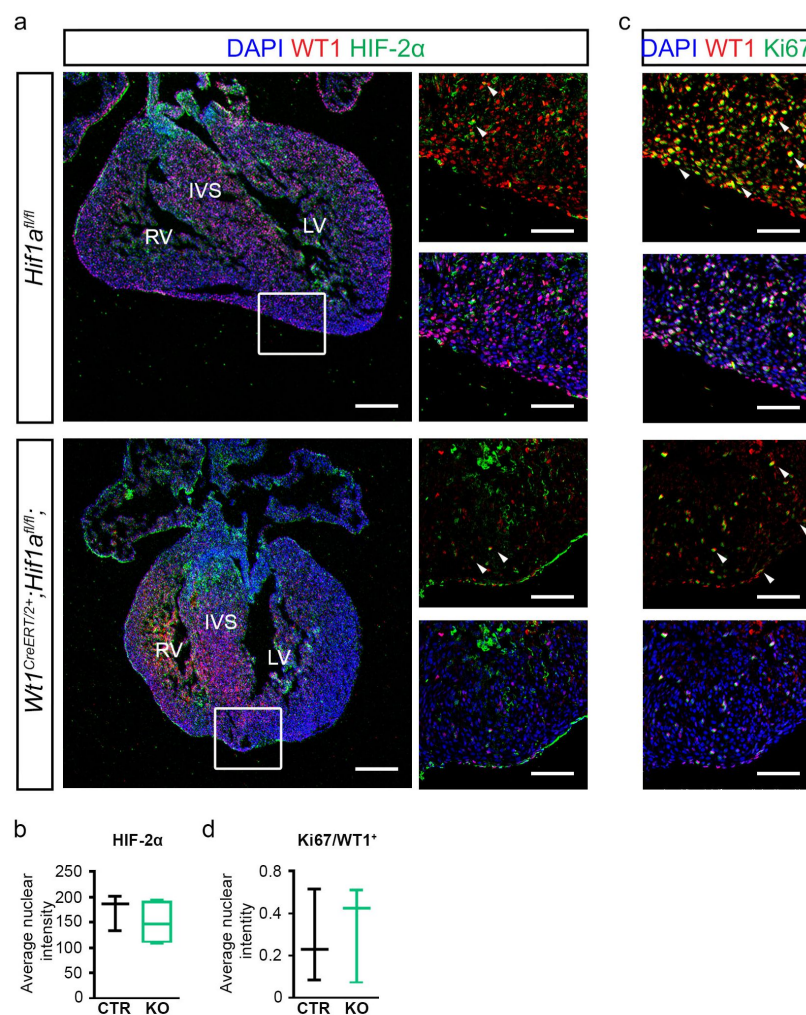


Figure 2-figure supplement 1.

Epicardial deletion of HIF-1α does not alter HIF-2α expression nor proliferation of WT1⁺ cells.

Representative images of co-immunostaining for WT1 (red) and HIF-2α (green) and DAPI nuclear stain (blue) and (b) quantification of HIF-2α nuclear signal in WT1+ cells on sections from *Hif1a^{fl/fl}* (CTR, $n = 3$) and *Wt1^{CreERT2/+};Hif1a^{fl/fl}* (KO, $n = 4$) hearts harvested at E16.5. (c) Representative images of co-immunostaining for WT1 (red), Ki67 (green) and DAPI nuclear stain (blue) and (d) quantification of Ki67 nuclear signal in WT1 + cells on section from CTR ($n = 3$) and KO ($n = 3$) hearts harvested at E16.5. Whole heart scale bars, 200μm; high magnification scale bars, 100μm. IVS= interventricular septum, LV= left ventricle, RV= right ventricle. Arrowheads indicate co-expression of WT1 and (a) HIF-2α or (b) Ki67. Data presented as median, IQR and upper and lower limits. n numbers refer to individual mice. Two-tailed, unpaired Student t-tests were used for statistical analysis.

Figure 2-figure supplement 2.

Tamoxifen treatment of *Wt1^{CreERT2/+};Hif1 α ^{fl/fl}*-derived epicardial explants reduces HIF-1 α expression without affecting HIF-2 α .

(a) Representative images of co-immunostaining for HIF-1 α (green), HIF-2 α (red) and DAPI nuclear stain (blue) and quantification of (b) HIF-1 α and (c) HIF-2 α nuclear signal from E11.5 derived epicardial explants of *Hif1 α ^{fl/fl}* (CTR, $n = 4$) and *Wt1^{CreERT2/+};Hif1 α ^{fl/fl}* (KO, $n = 4$). Scale bars, 100 μ m. Data presented as median, IQR and upper and lower limits. n numbers refer to individual mice. Two-tailed, unpaired Student t-tests were used for statistical analysis. * $p < 0.05$.

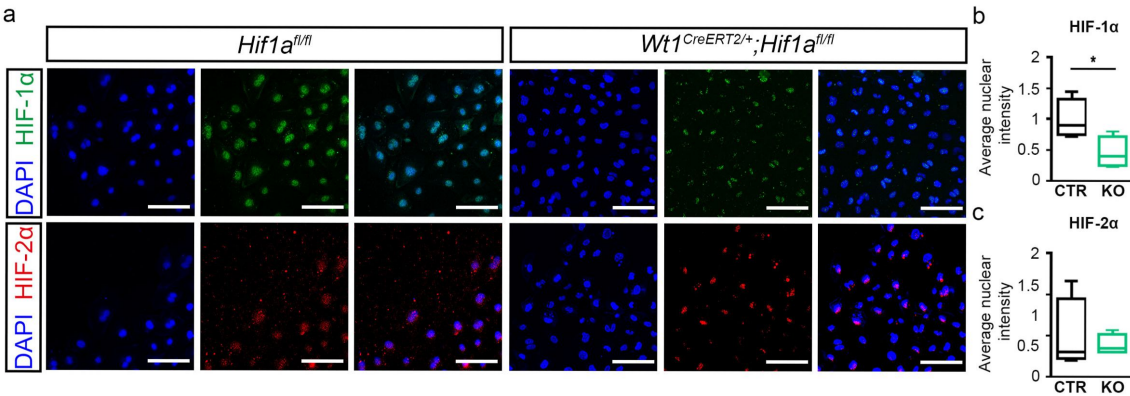
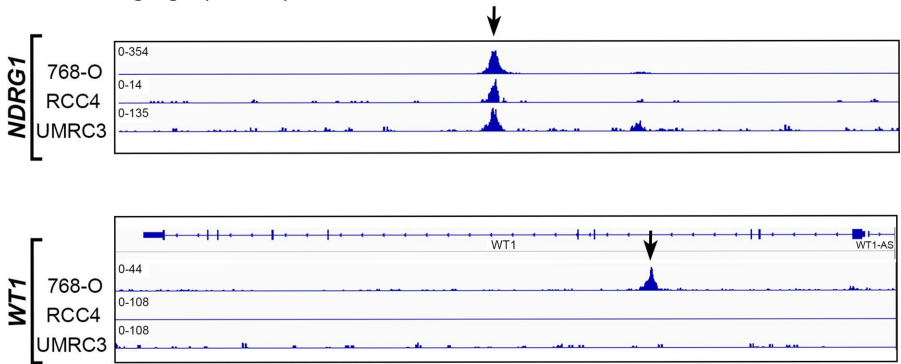


Figure 2-figure supplement 3.

HIF-1 β subunit directly binds to the WT1 locus.

HIF1 β ChIP-sequencing analysis of *NDRG1* (positive control), and *WT1* loci, conducted on human clear cell renal cell carcinoma (ccRCC) line 786-O. Arrows highlight positive peaks.



To elucidate the direct regulation of WT1 by HIF-1 α , we interrogated a publicly available dataset of chromatin immunoprecipitation coupled with next generation sequencing (ChIP-seq) generated in the renal adenocarcinoma 786-O cell line¹⁵. ChIP-seq analysis confirmed direct HIF binding within intron 3 of the *Wt1* gene (**Figure 2-figure supplement 3**), consistent with previous findings¹⁷.

In summary, epicardial-specific deletion of *Hif1a* *in vivo* and *ex vivo* significantly reduced the number of WT1+ cells, both in the epicardium and underlying myocardium, arising through EMT affecting coronary vessel development at E16.5. Taken together our data demonstrate that HIF regulates the expression of WT1 and associated epicardial EMT.

Stabilisation of HIF signalling enhances Wt1 expression and epicardial EMT

We next sought to investigate whether stabilising HIF signalling under normoxic conditions is sufficient to enhance epicardial EMT. To this end, we established epicardial explants from *Rosa26⁺/CreERT2*; *Phd2^{fl/fl}* mice, where Cre-mediated recombination induced loss of *Phd2*, thereby stabilising HIF signalling in all cell types but restricted to epicardium in explants for *ex vivo* analysis. Upon tamoxifen treatment, *Rosa26⁺/CreERT2*; *Phd2^{fl/fl}* derived explants (KO) exhibited a significant induction of HIF-1 α and HIF-2 α expression as compared to controls (CTR) (**Figure 3-figure supplement 1a-c**; HIF-1 α ; CTR: 0.26 ± 0.13 ; KO: 0.86 ± 0.09 ; $p = 0.0061$; HIF-2 α ; CTR: 0.37 ± 0.01 ; KO: 1.12 ± 0.15 ; $p = 0.0027$). Subsequent visualization of stress fibres via α -SMA staining revealed augmented mesenchymal morphology in the mutant-derived explants (**Figure 3a**). Consistent with this WT1 nuclear fluorescence was significantly increased in KO explants as compared to controls (CTR) (**Figure 3b-c**; CTR: 158.7 ± 48.34 ; KO: 330.5 ± 27.26 ; $p = 0.0364$). In addition, expression levels of the EMT transcription factor snail family transcriptional repressor 2 (*Snai2*) gene were increased in *Phd2*-KO explants relative to CTR (**Figure 3d**; CTR: 1 ± 0.025 ; KO: 1.166 ± 0.036 ; $p = 0.0464$).

To complement our genetic studies, we undertook a more therapeutically relevant approach by using a pharmacological PHD inhibitor to stabilise HIF signalling. Epicardial explants were generated from C57BL/6 wildtype embryos and treated with DMSO control (CTR) or Molidustat (BAY 85-3934, Mol) a specific PHD inhibitor, used in clinical trials for renal anaemia³⁰. Immunostaining for HIF-1 α and HIF-2 α confirmed the stabilisation of HIF signaling upon PHD inhibitor treatment (**Figure 3-figure supplement 1d-g**). Staining with phalloidin (**Figure 3e**) and α -SMA (**Figure 3f**) showed an enhanced mesenchymal morphology with increased stress fibres in inhibitor-treated explants, suggestive of EMT induction. Furthermore, immunostaining demonstrated a significant increase in WT1 expression in treated explants as compared to control (**Figure 3g-h**; CTR: 1039 ± 134.9 ; Mol: 2500 ± 503.7 ; $p = 0.03$). Likewise, the PHD inhibitor enhanced the motility of cultured mouse embryonic ventricular epicardial cells (MEC.1 cell line)³¹ compared to control-treated cells (**Figure 3i-j**; CTR: 436869 ± 9359 ; Mol: 372482 ± 20880 ; $p = 0.0477$), as assessed by scratch assays. These findings collectively suggest that genetic perturbation and/or pharmacological inhibition of PHDs stabilise HIF signalling in the epicardium *ex vivo*, leading to increased WT1 expression and stimulation of epicardial EMT.

Reduced HIF signalling is associated with epicardial quiescence in the neonatal heart

To gain unbiased insight into hypoxia and HIF-related molecular pathways which are differentially regulated in intact P1 versus P7 mouse hearts, we performed single cell RNA-seq analysis using the 10x Genomics Chromium platform and next generation sequencing. Cell clusters were visualised through uniform manifold approximation and projection (UMAP) (**Figure 4a**). An epicardial cell cluster (Epi) was identified based on previously described specific epicardial gene expression signatures³² (**Figure 4b**). As expected, a cardiomyocyte cell cluster (CM) was

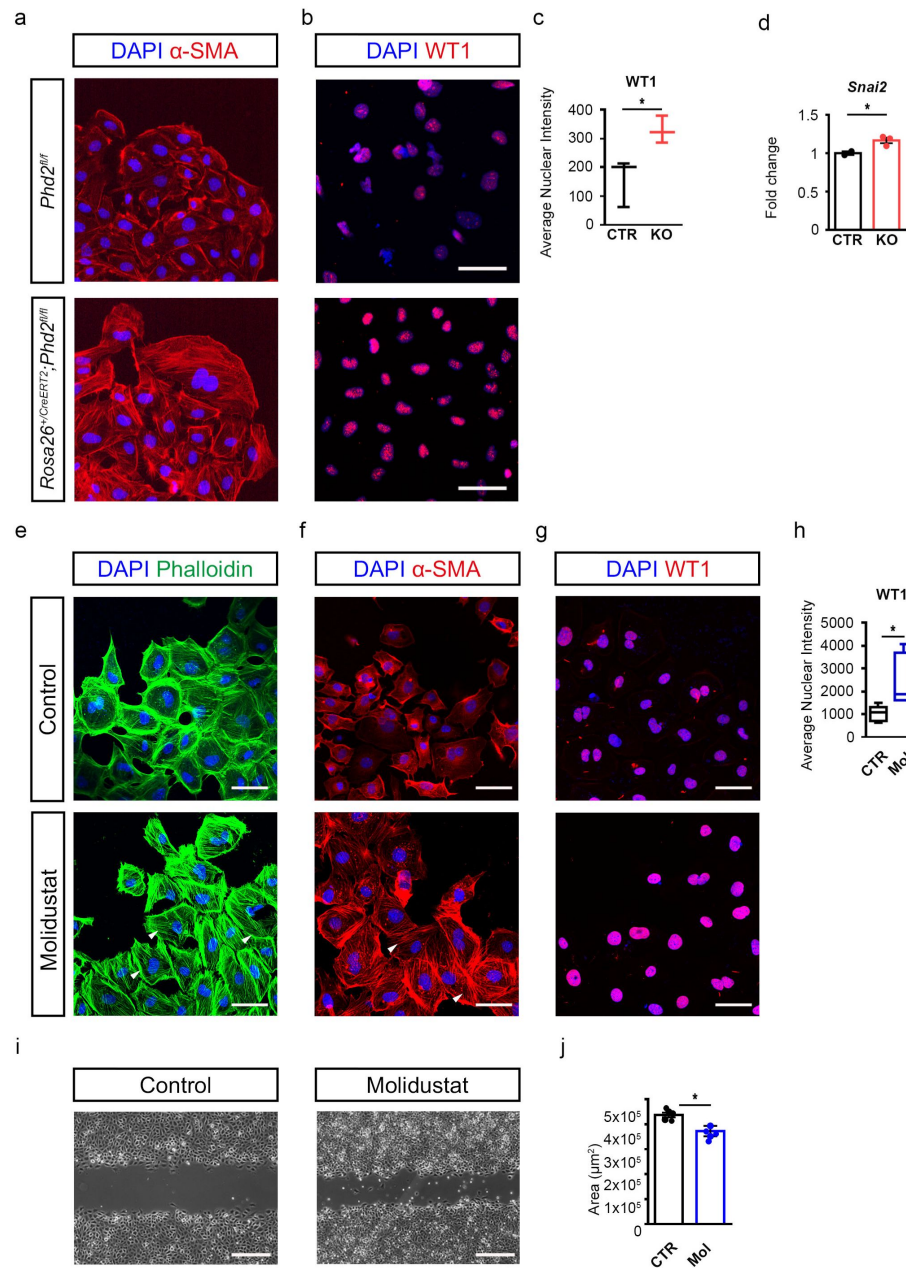


Figure 3.

Stabilisation of HIF signalling promotes Wt1 expression and enhances epicardial EMT.

(a,b) Representative images of immunostaining for α smooth muscle actin (α -SMA, red), WT1 (red) and DAPI nuclear stain (blue) and (c) quantification of nuclear intensity of WT1 signal on epicardial explants derived from *Phd2^{fl/fl}* (CTR) and *Rosa26^{+/CreERT2};Phd2^{fl/fl}* (KO) embryos at E11.5. $n = 5$ hearts/group. (d) *Snai2* expression analysis by qRT-PCR using RNA isolated from epicardial explants derived from CTR ($n = 2$) and KO ($n = 3$) hearts. (e-g) Representative images of immunostaining for Phalloidin (green), α -SMA (red), WT1 (red) and DAPI nuclear stain (blue), and (h) quantification of nuclear intensity of WT1 signal on epicardial explants derived from wild type hearts harvested at E11.5 and treated with DMSO (control, CTR, $n = 6$) or Molidustat (Mol, $n = 6$). Scale bars, 50 μm . Arrowheads indicate stress fibres. n numbers refer to individual mice. (i) Representative images and (j) quantification of cell migration assessed by wound healing/scratch assay of MEC.1 cells treated with DMSO (control, CTR, $n = 5$), or Molidustat (Mol, $n = 5$). n numbers refer to technical replicates. Scale bars, 0.5 μm . Data presented as median, inter-quartile range (IQR) and upper and lower limits or mean $\pm$ s.e.m. Two-tailed, unpaired Student t-tests were used for statistical analysis. * $p < 0.05$.

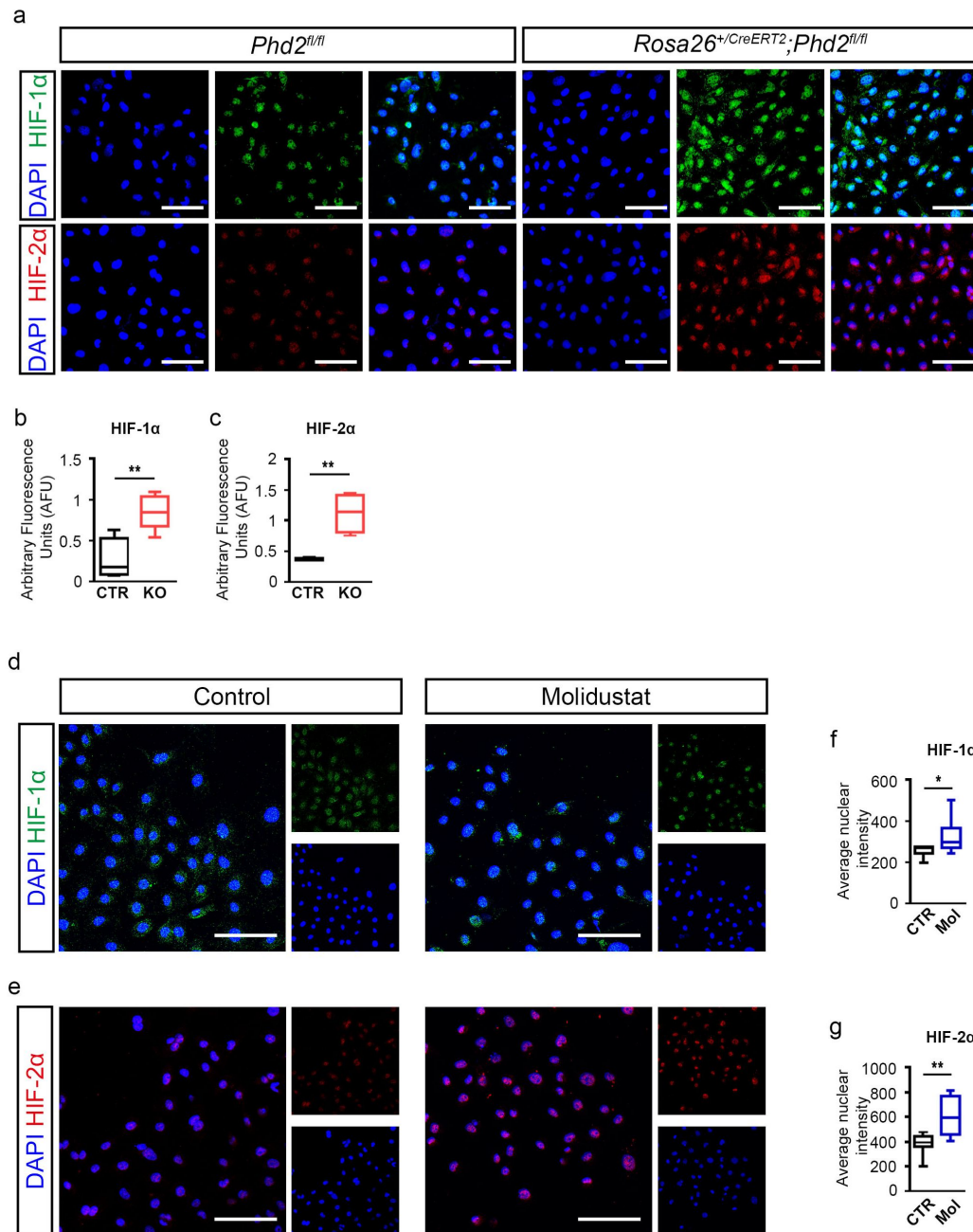


Figure 3-figure supplement 1.

Genetic and pharmacological stabilisation of HIF signaling induces both HIF-1α and HIF-2α expression in epicardial explants.

(a) Representative images of co-immunostaining for HIF-1α (green), HIF-2α (red) and DAPI nuclear stain (blue) on epicardial explants derived from E11.5 hearts of *Phd2^{fl/fl}* (CTR) and *Rosa26⁺/CreERT2⁺;Phd2^{fl/fl}* (KO) embryos. Quantification of (c) HIF-1α and (d) HIF-2α nuclear signal. N=4/group. (d,e) Representative images of co-immunostaining for HIF-1α (green), HIF-2α (red) and DAPI nuclear stain (blue) on epicardial explants derived from E11.5 hearts of wild type embryos treated with DMSO (Control), or Molidustat (BAY 85-3934). Quantification of (f) HIF-1α and (g) HIF-2α nuclear signal from control (CTR, *n* = 4) and Molidustat (Mol, *n* = 4). Scale bars, 100μm. Data presented as median, IQR and upper and lower limits. *n* numbers refer to individual mice. Two-tailed, unpaired Student t-tests were used for statistical analysis. **p*<0.05, ***p*<0.01.

underrepresented due to both low survival and size incompatibility with the FACS cell sorting³³. Gene Ontology (GO) analysis for biological processes showed an enrichment of hypoxia-related pathways in P1 compared to P7 hearts (**Figure 4-figure supplement 1a**), particularly in the epicardial cell population (**Figure 4c**). In contrast, hypoxia-related pathways were enriched in P7 hearts compared to P1 hearts in the fibroblast cluster (**Figure 4-figure supplement 1b-d**), suggesting a role for hypoxia post-injury, specifically in the fibrotic response leading to scarring. Accordingly, expression of well-known HIF-induced genes such as *Vegfa* and pyruvate dehydrogenase kinase 3 (*Pdk3*) was increased in P1 derived epicardial cells (**Figure 4d**). Conversely, the expression of *Phd2*, encoding for the main suppressor of HIF signalling, was enriched in P7 cells (**Figure 4e**). These findings were further confirmed by a time-course analysis of mRNA levels using ventricle lysates and immunostaining which revealed a marked upregulation of *Phd2* cardiac levels from P7 to adulthood (**Figure 4f**, **Figure 4-figure supplement 2a-b**). Notably, this increase in *Phd2* levels coincided with a reduction in WT1 expression in P7 versus P1 hearts (**Figure 4g**).

Activation of HIF signalling improves the response to injury in non-regenerative P7 hearts

To establish a causative role in the regulation of epicardial WT1 activity, we extended HIF signaling beyond P7 using a genetic gain-of-function approach. Neonatal *Wt1^{CreERT2/+};Phd2^{fl/fl}* pups received a single dose of tamoxifen at P2 to induce Cre-mediated loss of *Phd2*. MI was then induced by permanent ligation of the proximal left anterior descending (LAD) coronary artery in P7 mice (**Figure 5a**), a time-point when the heart's regenerative capacity is lost. Hearts were collected at 21 dpi. Immunostaining analysis of the endothelial marker CD31 and α -SMA did not reveal a significant difference in the number of vessels between *Phd2^{fl/fl}* (CTR) and *Wt1^{CreERT2};Phd2^{fl/fl}* (KO) hearts (**Figure 5b-c**, CTR 0.0030 ± 0.0004 , KO 0.0034 ± 0.0002 , $p = 0.45$), suggesting that activation of HIF signalling in WT1-expressing cells does not affect the neovascularisation response. To investigate whether epicardial activation of HIF signalling improved heart function following MI, we performed cine-MRI analysis at 3-weeks post-MI. We observed a substantial increase in the ejection fraction (EF) and decrease in the end-diastolic volume (EDV) in KO animals compared to controls (**Figure 5d-f**; EF: CTR 43.21 ± 6.25 , KO 64.30 ± 6.82 , $p < 0.05$. EDV: CTR 54.19 ± 3.50 , KO 33.72 ± 2.67 ; $p < 0.05$). Furthermore, histological analysis by Masson's trichrome staining revealed a significant reduction of the fibrotic scar in KO animals as compared to controls (**Figure 5g-h**, CTR 15.44 ± 1.088 , KO 10.34 ± 1.374 , $p = 0.027$) without changes in the size of cardiomyocytes, as assessed by wheat germ agglutinin staining (WGA) (**Figure 5i-j**, CTR 167.6 ± 3.27 , KO 179.3 ± 8.68 , $p = 0.32$). Taken together these findings suggest that activated hypoxia signalling in the epicardium contributes to heart regeneration by reducing scar size and improving heart function.

To further investigate our hypothesis that prolonged maintenance of HIF signalling has a beneficial effect on the epicardium and response to MI in the postnatal heart, we explored a complementary pharmacological approach recapitulating our gain of HIF-function studies using epicardial explants. We induced MI by LAD ligation in P7 mice and administered Molidustat or DMSO control (vehicle) by intraperitoneal injection, immediately after surgery and one week later, to stabilise HIF signalling in the postnatal heart (**Figure 6a**). Treatment with PHD inhibitor effectively stabilized HIF-1 α and HIF-2 α , as observed by immunostaining (**Figure 6-figure supplement 1a-b**) in hearts collected at 9-days post-MI (9dpi). Moreover, this pharmacological approach revealed a significant increase in epicardial WT1 expression in both the infarct and remote zones (**Figure 6b**) in Molidustat-treated hearts (MI T) compared to controls (CTR) (**Figure 6c-d**, infarct zone: MI CTR 0.34 ± 0.08 , MI T 0.50 ± 0.06 , remote zone: CTR 0.075 ± 0.02 , MI T 0.25 ± 0.05 , $p < 0.05$). Similarly, an increase in WT1 expression in the myocardium was observed in the PHD-inhibitor treated group and was more pronounced proximal to the infarct region (**Figure 6e-f**, infarct zone: CTR 5.02 ± 1.34 , MI T 7.09 ± 1.86 , $p < 0.05$; remote zone: CTR 13.88 ± 4.96 , MI T

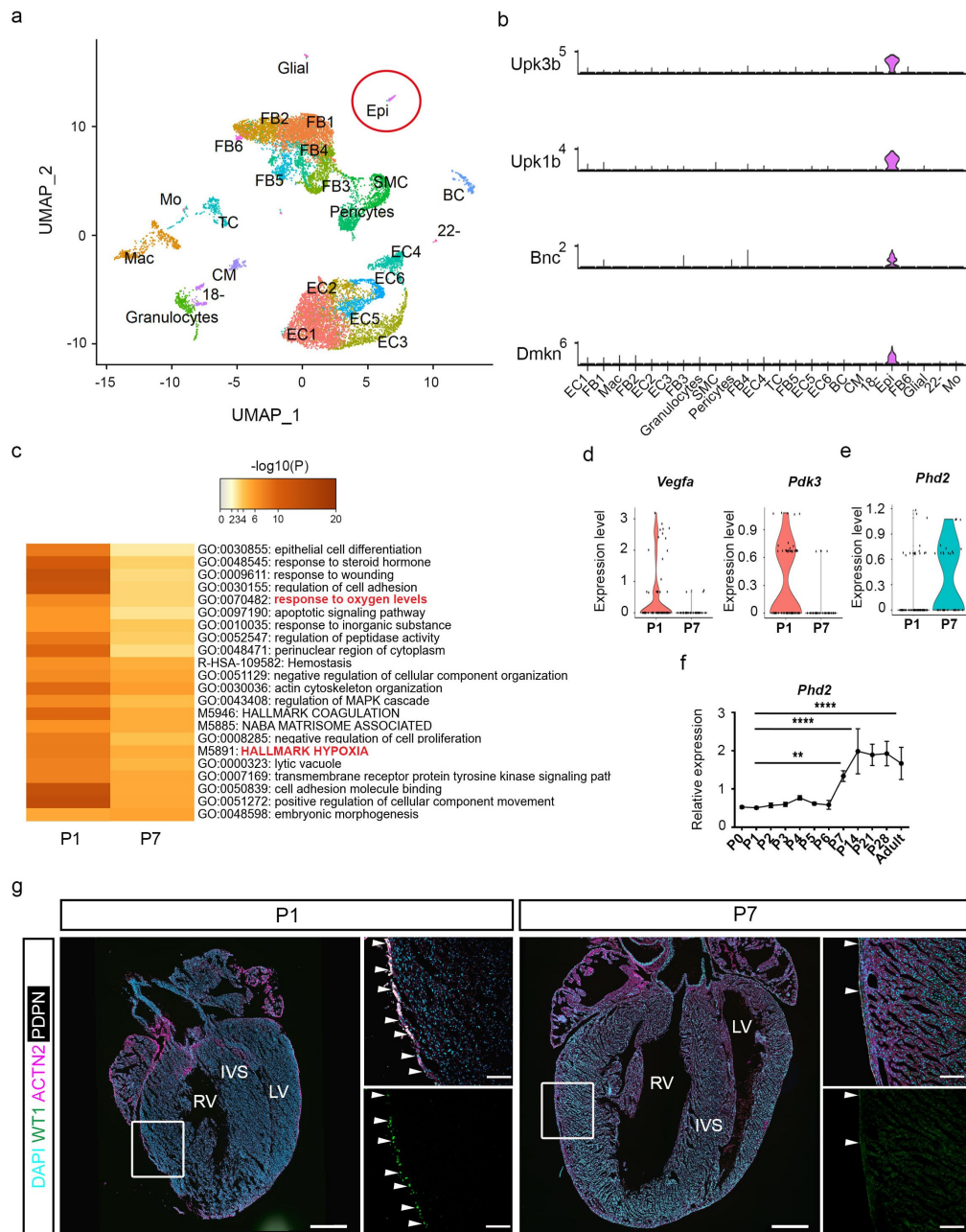


Figure 4.

HIF signalling is downregulated in between P1 and P7 in neonatal mice.

(a) UMAP representation of different cell populations in the intact neonatal heart at postnatal day (P) 1 and P7. (b) Stacked violin plots showing expression of canonical Epi-enriched genes *Upk3b*, *Upk1b*, *Bnc*, *Dmkn*. (c) Heatmap showing biological processes enriched in the Epi cluster at P1 versus P7. Violin plots showing expression of HIF target genes (d) *Vegfa* and *Pdk3*, and (e) *Phd2* at P1 versus P7. (f) Real time RT-PCR analysis of *Phd2*, $n = 4$ hearts per group. (g) Representative images of immunostaining for WT1 (green), ACTN2 (magenta), PDPN (white) and DAPI nuclear stain (cyan) at P1 and P7. Images to the right are magnified views of boxed regions shown in whole heart images. Arrowheads indicate expression of WT1 in the epicardium. IVS= interventricular septum, LV= left ventricle, RV= right ventricle. Whole heart scale bars, 200 μ m; high magnification scale bars, 100 μ m. $n = 4$ hearts per stage. Data are presented as mean $\pm$ s.e.m. n numbers refer to individual mice. One-way ANOVA with Tukey's post-hoc tests were used for statistical analysis. ** $p < 0.01$; **** $p < 0.0001$. Epi, epicardium.

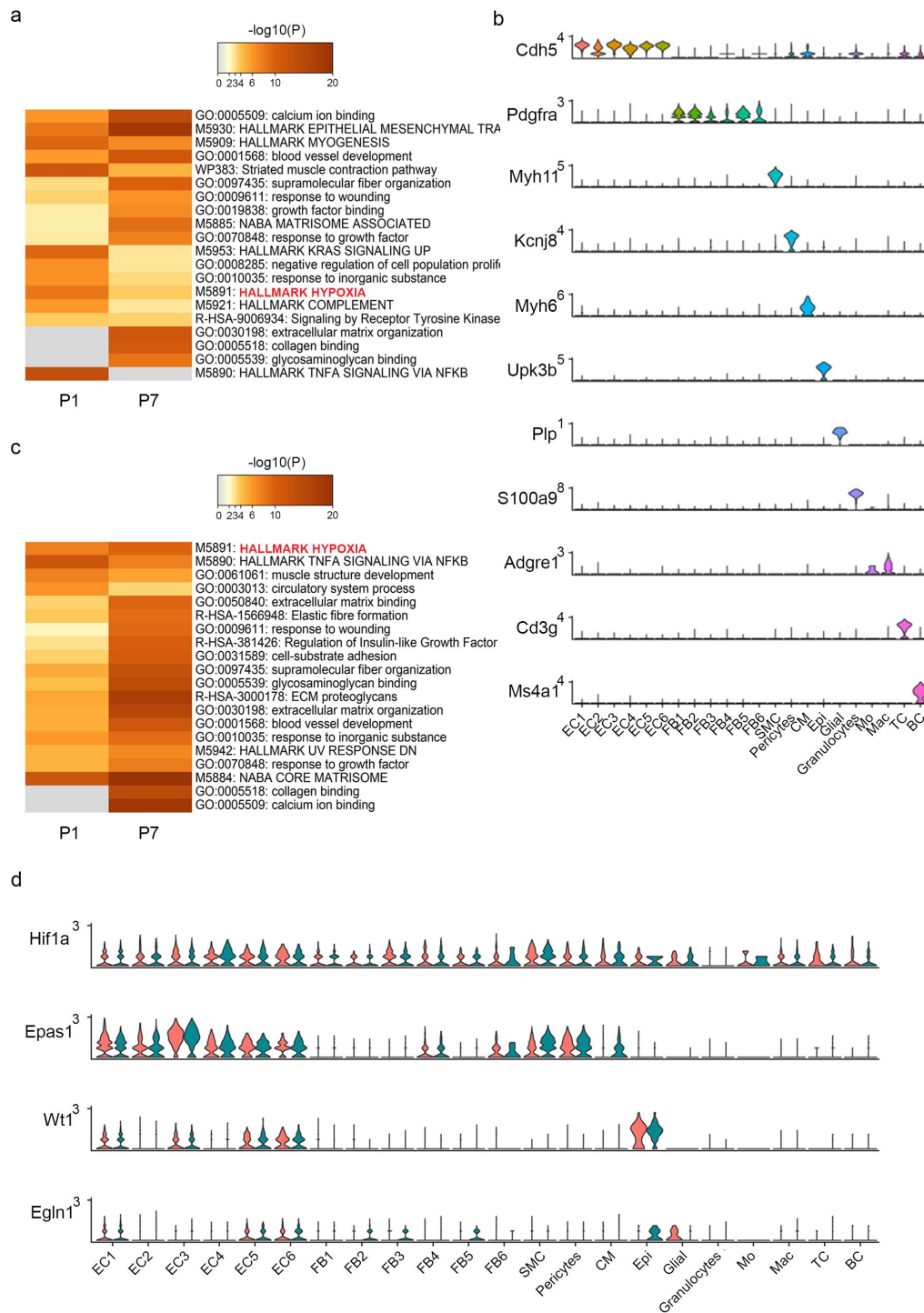


Figure 4-figure supplement 1.

Differential activity of HIF signalling in the neonatal mouse.

(a) Heatmap showing biological processes enriched in all cell populations from intact P1 and P7 hearts. (b) Stacked violin plots showing expression of canonical enriched genes for each cell cluster. (c) Heatmap showing biological processes enriched in the fibroblast cluster at P1 and P7. (d) Stacked violin plots showing expression of *Hif1a*, *Epas1* or *Hif2a*, *Wt1* and *Egl1* or *Phd2* genes for each cell cluster. Red violin represents P1 derived cells, green violin represents P7 derived cells. EC, endothelial cells; FB, fibroblast; SMC, smooth muscle cells; CM, cardiomyocyte; Epi, epicardium; Mo, monocytes; Mac, macrophages; TC, T-cells; BC, B-cells.

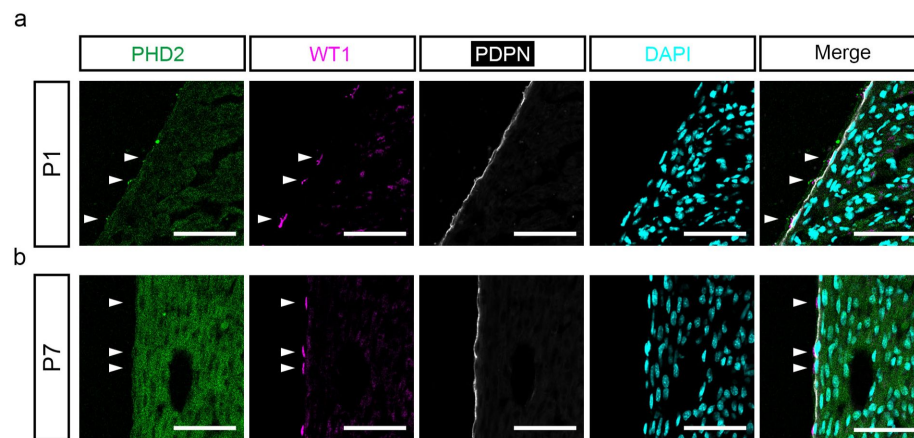


Figure 4-figure supplement 2.

PHD2 expression increases in the neonatal heart from P1 to P7.

Representative images of immunostaining for PHD2 (green), WT1 (magenta), PDPN (grey) and DAPI (cyan) on cryosections of postnatal (right) ventricles of P1 (a) and P7 hearts (b). Arrowheads indicate PHD2 expression in WT1⁺ cells. *n*=3 hearts per stage. Scale bar, 50μm.

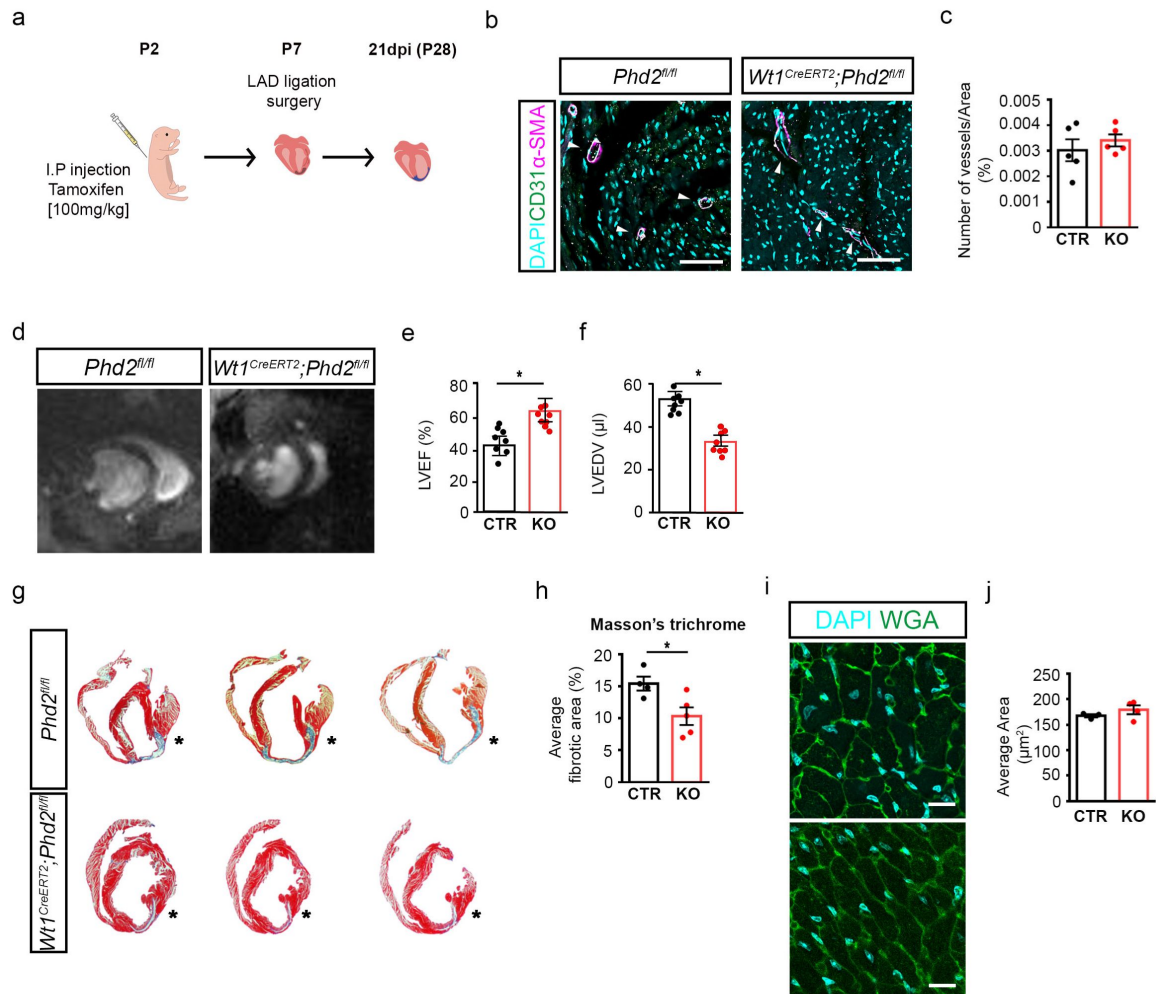


Figure 5.

Loss of *Phd2* in WT1 expressing cells improves heart regeneration post-MI.

(a) Schematic of experimental design. (b) Representative images of immunostaining for α -smooth muscle actin (α -SMA, magenta), CD31 (green) and DAPI nuclear stain (cyan) on sections of hearts from *Phd2^{fl/fl}* (CTR, $n = 5$) and *Wt1^{CreERT2};Phd2^{fl/fl}* (KO, $n = 5$) mice and (c) quantification of number of vessels at 21 days post-injury (dpi). Arrowheads indicate vessels (determined by co-expression of CD31 and α -SMA). Scale bars, 100 μ m. (d) Representative mid-ventricular short-axis MRI frames for *Phd2^{fl/fl}* (CTR) and *Wt1^{CreERT2};Phd2^{fl/fl}* (KO) mice hearts. (e-f) MRI analyses of infarcted hearts at 21 dpi showing increased EF (e) and reduced EDV (f) in *Wt1^{CreERT2};Phd2^{fl/fl}* (KO, $n = 8$) animals compared with controls (CTR, $n = 8$). (g) Representative images and (h) quantification of Masson's Trichrome stained transverse serial sections to assess cardiac fibrosis (blue) at 21dpi. * denotes suture placement. (i) Representative images of immunostaining for wheat-germ agglutinin (WGA, green) and DAPI nuclear stain (cyan) on hearts from control and Molidustat treated mice 21dpi for: Scale bars, 100 μ m. (j) Quantification of cardiomyocyte cell size, as assessed by WGA staining on sections of hearts from CTR ($n = 4$) and KO ($n = 4$). Left ventricular regions were measured. Error bars represent mean $\pm$ s.e.m. n numbers refer to individual mice. Two-tailed, unpaired Student t-tests were used for statistical analysis. * $p < 0.05$.

18.61 ± 6.18; $p < 0.05$). We observed an extensive co-localization of WT1 and EMCN within the myocardial compartment at 9dpi (**Figure 6-figure supplement 1c**), supporting an endothelial identity for these cells. Furthermore, an assessment of the proliferation of WT1⁺ cells and cardiomyocytes revealed no significant differences between control and Molidustat-treated animals (**Figure 6-figure supplement 1d, e, g**).

To analyse the long-term histological effects of HIF-stabilisation after MI, we collected hearts at 21dpi. Co-immunostaining and quantification of CD31 and α -SMA revealed no difference in the number of vessels between controls and the treatment group (**Figure 6g-h**). Cine-MRI analysis showed a marked reduction in left ventricular systolic function following MI in the vehicle-treated group (MI CTR) as compared to uninjured hearts (Sh) and, more importantly, a significant improvement in systolic function in the PHD inhibitor treated-groups (MI T). EF and EDV were significantly improved in the drug-treated groups (**Figure 6i-j**; EF: Sh 78.10 ± 3.22, MI CTR 67.70 ± 4.31, MI T 77.61 ± 1.76, $p < 0.05$; EDV: Sh 27.53 ± 3.10, MI CTR 36.11 ± 2.49 MI T 26.44 ± 0.93; $p < 0.05$). Histological analysis by Masson's trichrome staining revealed the presence of a fibrotic scar in drug-treated animals, albeit this was reduced relative to controls (**Figure 6k-l**) and the at-risk myocardium was comparatively thicker with minimal dilatation or thinning of the left ventricle. The thickened myocardium was not due to an effect on cardiomyocyte hypertrophy, as determined by WGA staining (**Figure 6-figure supplement 1f-h**) nor an effect on cardiomyocyte proliferation (**Figure 6-figure supplement 1e, g**) suggesting potential cardio-protection and improved preservation of survived heart muscle. More generally these findings also indicate that improved remodelling and function following drug treatment are possible against a background of fibrosis.

Discussion

In this study, we characterised the levels of hypoxia in the heart during mid-to-late gestation when the epicardium is fully formed. At E12.5, hypoxia was mainly observed in areas of compact myocardium, most significantly in the atrioventricular (AV) groove, close to where the coronary plexus first develops³⁴. By E14.5, hypoxic regions localised to the interventricular septum (IVS), which is perfused later than the muscle of the free wall³⁵. Finally, by E16.5 and E18.5, when the heart is adequately perfused by the coronary vessels, hypoxia was primarily restricted to the epicardium. We further analysed the expression and distribution analysis of HIF-1 isoforms, finding a predominance of the HIF-1 α isoform in WT1⁺ cells, whereas HIF-2 α was more localized in the myocardium. Given hypoxia's known role in promoting EMT³⁶ we confirmed a functional role for HIF-1 α in promoting morphological changes associated with EMT by direct regulation of *Wt1* expression (**Figure 2** and **3**, **Supplementary Figure 5**). Moreover, epicardial-specific deletion of *Hif1a* significantly reduced the number of WT1 expressing cells in both epicardial and myocardial compartments, along with impaired coronary vessel development in E16.5 embryos. Collectively, this suggests that HIF-1 α induces epicardial EMT and plays an essential role in coronary vessel formation during development. While Cre-mediated deletion of HIF-1 α , induced at E9.5 and E10.5, predominantly targeted epicardial cells, it is worth noting that expression of WT1 in the coronary endothelium has been reported as early as E1 1.5³⁷, although the biological significance of endothelial WT1 remains elusive. Thus, aberrant HIF signalling in WT1⁺ endothelial cells cannot be completely excluded as contributing to the observed defects on the expansion of the coronary vasculature. However, stabilisation of HIF signalling in epicardial explant cultures, under physiological oxygen levels, either via genetic ablation of *Phd2*, or chemical inhibition of PHD enzymes with Molidustat, resulted in enhanced EMT and WT1 expression, further confirming a role for HIF signalling in epicardial activation.

To gain insight into the post-natal molecular pathways regulating the epicardium, we performed single cell RNA sequencing analysis comparing P1 and P7 stages, focusing on the specific gene expression signature of the epicardial cell cluster. GO terms analysis showed an enrichment of

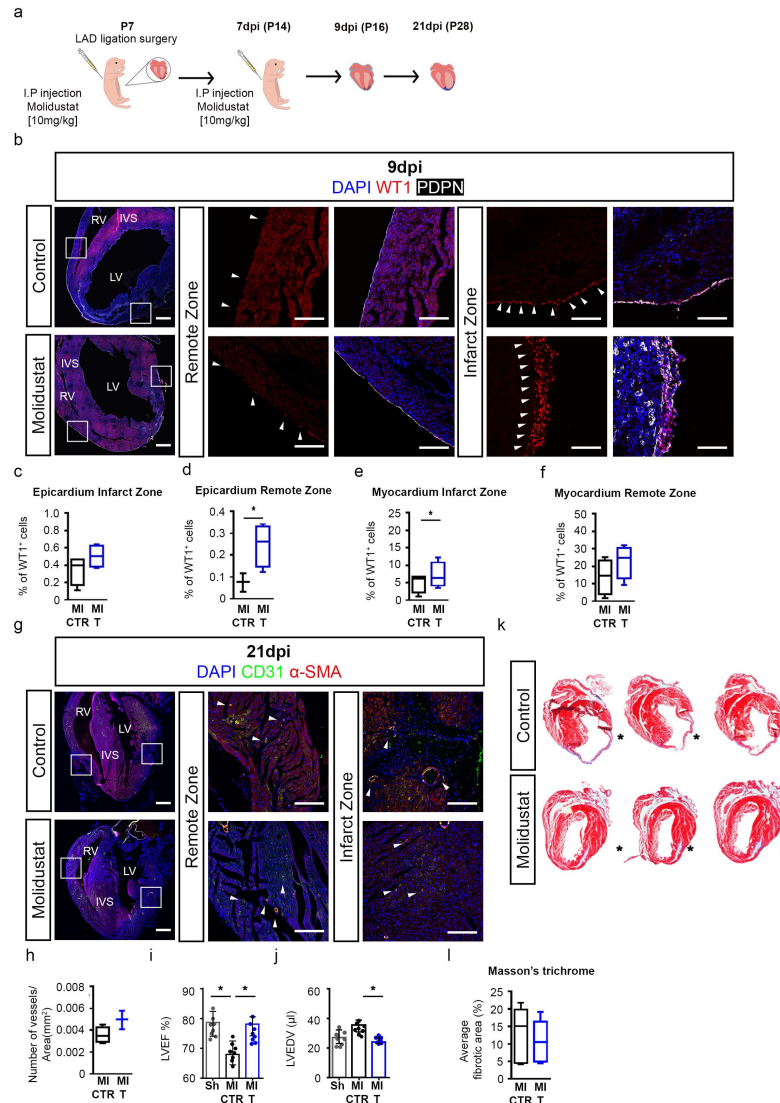


Figure 6.

Pharmacological treatment with PHD inhibitors induces WT1 expression and improves heart function.

(a) Schematic of experimental design. (b) Representative images of immunostaining for WT1 (red), PDPN (white) and DAPI nuclear stain (blue) on sections of hearts from animals treated with either saline (MI CTR) or Molidustat (MI T) and (c-f) quantification of number of WT1 positive cells at 9 days post-injury (dpi). Images on the right represent high magnification views of boxed regions shown in whole heart images. Arrowheads indicate expression of WT1 in the epicardium (determined by co-expression with PDPN). $n = 4$ animals per group. (g) Representative images of immunostaining for CD31 (green), α -SMA (red) and DAPI (blue) on sections of hearts from animals treated with either saline (MI CTR, $n = 4$) or Molidustat (MI T, $n = 4$) and (h) quantification of number of vessels at 21dpi. Arrowheads indicate vessel (determined by co-expression of CD31 and α -SMA). Whole heart scale bars, 200 μ m; high magnification scale bars, 100 μ m. Left ventricle (i) Ejection fraction (LVEF) and (j) end diastolic volume (LVEDV) evaluated by MRI analyses of non-infarcted (Sham, Sh; $n = 8$), infarcted (MI CTR; $n = 8$) controls and treated (MI T; $n = 8$) hearts at 21dpi. (k) Representative images and (l) quantification of Masson's Trichrome stained transverse sections to assess cardiac fibrosis (blue) at 21dpi. * denotes suture placement. Scale bars, 1mm. IVS= interventricular septum, LV= left ventricle, RV= right ventricle. Scale bars 200 μ m, high magnification scale bars, 50 μ m. Data presented as median, IQR and upper and lower limits and as mean $\pm$ s.e.m. n numbers refer to individual mice. Two-tailed, unpaired Student t -tests were used for statistical analysis between two groups, one-way ANOVA with Dunnett's post-hoc test was used for multiple comparisons in i and j. * $p < 0.05$.

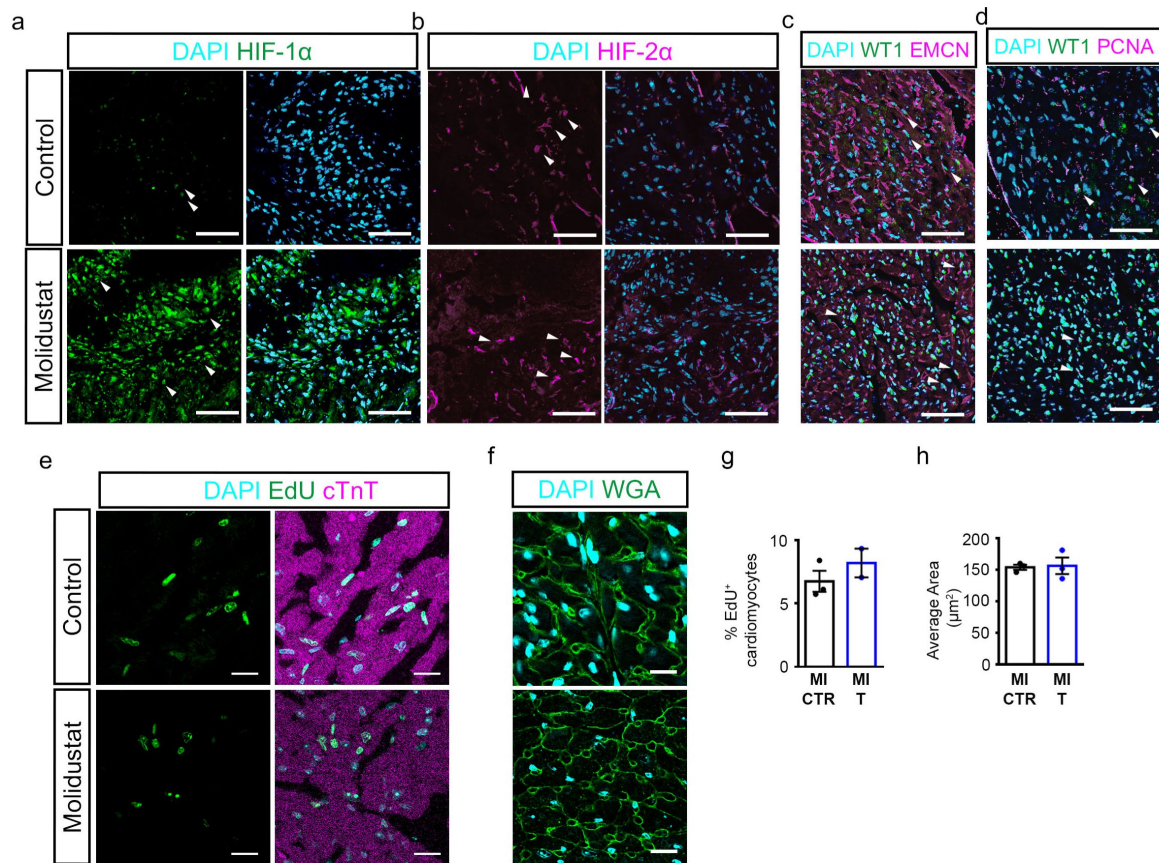


Figure 6-figure supplement 1.

Effects of PHD inhibitor Molidustat administration to infarcted neonatal hearts at P7 and P14 on HIF signaling, cell proliferation and hypertrophy.

Representative images of immunostaining on cryosections of hearts from control and Molidustat treated mice 9dpi for: (a) HIF-1 α (green), (b) HIF-2 α (magenta), (c) WT1 (green) and EMCN (magenta), (d) WT1 (green) and PCNA (magenta), (e) EdU (green) and cardiac troponin T (cTnT, magenta), (f) wheat-germ agglutinin (WGA, green) and DAPI nuclear stain (cyan). Arrowheads indicate area of expression or co-expressing WT1/EMCN or WT1/PCNA. Scale bars, 100 μ m. (g) Quantification of EdU positive cardiomyocytes expressed as percentage of total cardiomyocytes in hearts from control (MI CTR, $n = 3$) and Molidustat (MI T, $n = 2$) treated mice. (h) Quantification of cardiomyocyte cell size, as assessed by WGA staining on sections of hearts from control (MI CTR, $n = 3$) and Molidustat (MI T, $n = 3$) treated mice. Left ventricular regions were measured. Error bars represent mean $\pm$ s.e.m. n numbers refer to individual mice. Two-tailed, unpaired Student t-tests were used for statistical analysis.

hypoxia-related pathways in P1 hearts, consistent with the expression of well-known HIF target genes. Importantly, *Phd2* expression was enriched in P7 hearts, potentially contributing to epicardial quiescence during the first week of life, concurrent with a decrease in HIF signalling. Complete regeneration following MI requires both the replacement of lost cardiomyocytes and the formation of new blood vessels. Given the role of the epicardium during development in promoting coronary vessel formation and myocardial growth, the observed epicardial quiescence after birth likely contributes to the loss of regenerative capacity. Genetic and pharmacological stabilisation of HIF signalling beyond P7 proved to be effective in extending the regenerative window after LAD surgery, resulting in reduced pathological remodelling and preserved cardiac function against a background of fibrotic repair. Interestingly, genetic stabilisation of HIF signalling specifically in WT1-expressing cells significantly reduced scar size post-injury (**Figure 5**). Reduced fibrosis was also observed in PHD-inhibitor treated P7 mice (**Figure 6**). This is important since necrotic cardiomyocytes can trigger the activation of fibroblasts and promote fibrosis³⁸. The epicardium is a key source of mitogens in the embryonic heart³⁹ and provides paracrine signals targeting cardiomyocytes during regeneration^{39,40}. In our study, we did not observe any change in either cardiomyocyte proliferation or hypertrophy, strengthening the hypothesis of an epicardium-mediated role in cardiomyocyte protection from apoptosis via paracrine signalling⁷.

It is noteworthy that despite persistent scarring (albeit reduced) following either genetic or pharmacological perturbation of PHDs, significant functional improvements were observed, in contrast to the prevailing view that fibrotic repair after MI is a barrier to effective tissue regeneration^{41–43}. Previous studies have suggested the involvement of HIF-1 α in myocardial remodelling after injury. Mice with constitutive overexpression of HIF-1 α in the myocardium showed enhanced angiogenesis, attenuation of infarct size, and improved cardiac performance after MI⁴⁴. Furthermore, cardiac-specific PHD2 inactivation and consequent HIF activation play a causal role in the pathogenesis of ischemic cardiomyopathy⁴⁵. In our study, we focused on the stabilisation of HIF signaling as a therapeutic approach to improve outcomes after heart injury. The mean half-life of Molidustat ranges from 4 to 10 hours⁴⁶ ensuring prolonged stabilisation of HIF signaling to enable therapeutic benefit. Molidustat is an orally administered small molecule used for the treatment of anaemia in patients with non-dialysis-dependent chronic kidney disease (CKD)³⁰ and could potentially be repurposed to treat ischemic heart disease.

In summary, we show that the epicardium is hypoxic at later stages of development, with HIF-1 α playing a crucial role in epicardial activation and EMT, necessary for supporting coronary vessel development. The epicardium becomes quiescent after birth, coinciding with decreased HIF-signalling. However, maintaining HIF signaling after birth can extend epicardial activation, leading to a significant improvement in cardiac remodelling and function after injury, thus representing a potential novel therapeutic target to improve cardiac regeneration following MI.

Materials & Methods

Mouse Lines

All mice were housed in individually ventilated cages (IVCs) and ventilated racks at 22°C and 55% humidity, under controlled environmental conditions according to the United Kingdom Home Office. The following mouse lines were used: Rosa²⁶⁺/CreERT2;*Hif1 α* ^{fl/fl}, Rosa²⁶⁺/CreERT2;*Phd2*^{fl/fl}. The *Wt1*^{CreERT2/+}; *Hif1 α* ^{fl/fl} and *Wt1*^{CreERT2/+}; *Phd2*^{fl/fl} mouse line was generated by crossing *Wt1*^{CreERT2/+}²⁰ with *Hif1 α* ^{fl/fl} animals or *Phd2*^{fl/fl} for two generations. Genetically modified mouse lines used were kept in a pure C57BL/6 background. Both males and females were used in the study. For timed-mating experiments, 8-12-week-old mice were set up overnight and females checked for vaginal plugs the following morning; the date of a vaginal plug was set as embryonic day (E) 0.5. For tamoxifen-dependent gene activation, 2 doses of 40 mg/kg of body weight of

tamoxifen (Sigma) were administered to pregnant dams by oral gavage, at embryonic stages E9.5 and E10.5. For neonate studies, pups were injected intraperitoneally (i.p.) with a single 10 μ l dose of 20 mg/ml tamoxifen, at postnatal day (P)2, using a 25-gauge needle. 5-ethynyl-2'-deoxyuridine (EdU, Thermofisher) was intraperitoneally injected at a dose of 50mg/kg at the day of surgery (P7) and every other day. Hearts were collected at 9 dpi and the assay was carried out according to manufacturer's protocol.

Hypoxyprobe

For hypoxia studies, pregnant females were injected i.p. with 1.5 mg Hypoxyprobe (pimonidazole hydrochloride, Hypoxyprobe-1 Inc). After 2 hours, hearts from embryos were harvested and fixed. Pimonidazole is a 2-nitroimidazole that is reductively activated specifically in hypoxic cells and forms stable adducts with thiol groups in proteins, peptides, and amino acids at oxygen levels below 1.3%.

Cell lines

The mouse immortalised embryonic epicardial cell line MEC.1²⁶ was purchased from Merck/Millipore (SCC187) and cultured according to manufacturer' recommendations.

Myocardial Infarction

All procedures were carried out following local ethical approval (AWERB) at the University of Oxford and under the regulation of a United Kingdom Home Office project licences (references PPL30/2987, PPL30/3155, PDDE89C84 and PP3194787) in full compliance with the Animals (Scientific Procedures) Act 1986 (A(SP)A, revised 2012). Animals were euthanized according to the UK Home Office designated schedule 1 method of cervical dislocation. Myocardial infarction (MI) was induced by permanent ligation of the left anterior descending (LAD) coronary artery in wild type mice at postnatal day (P) 7, as previously described⁴⁷. All surgery was performed under isoflurane anaesthesia, and every effort was made to minimise suffering. Neonates cannot be sexed during the first week of life so mixed male and female cohorts were used throughout the protocol. Mice were anesthetized with 4% isoflurane for 1 minute and then transferred to an ice box for up to 6 minutes. Mice were recovered under 0.5% isoflurane. Animals received intraperitoneal injections of BAY 85-3934 (Molidustat, Selleck) (10 mg /kg) or vehicle (DMSO), upon recovery (day 0) and 7 days later. Mice were sacrificed by cervical dislocation for tissue collection at 4-, 9- and 21-days following ligation. Mice with excessive or minimal infarct sizes at early time points were excluded from further histological or functional assessment (see under Cine MRI). Investigators were blinded to mouse genotype and treatments prior to sample analyses.

Epicardial explants

Sterile 12-well plates (Fisher Scientific), containing sterile 13-mm diameter coverslips were coated with 0.1% gelatin (Millipore) and allowed to stand for 20 minutes. The gelatin was then replaced by Dulbecco's Modified Eagle Medium (DMEM) (Sigma) containing 10% FBS (Sigma), 1% Penicillin/Streptomycin (Sigma). Hearts were isolated from E11.5 embryos and the outflow tract and atria were removed. Each ventricle was then cut in half and placed epicardial side down on the coverslip. Explants were kept at 37°C/5% CO₂. 24 hours later, 1 μ M of tamoxifen was added to explants. After 48 hours of culture, the explant tissue was carefully peeled off, leaving only the epicardial sheet remaining on the coverslip. For explants prepared from C57BL/6 embryos, 50 μ M of BAY 85-3934 (Molidustat, Selleck) or vehicle (DMSO) was added. After a total of 72 hours in culture, the media was removed, coverslips were washed briefly in ice-cold PBS and then fixed in 4% PFA at room temperature for 15 minutes. The MEC.1 cell line identity was authenticated by qPCR and immunostaining for known epicardial markers and tested negative for mycoplasma contamination. Coverslips were subsequently washed in PBS before proceeding with immunostaining.

Histological analysis

Following overnight (O/N) fixation in 2% PFA at 4°C, hearts were washed in PBS and dehydrated by passage through rising concentrations of ethanol. Samples were then washed in Butanol O/N, at RT, before being placed in molten 50:50 butanol: Histoplast paraffin wax (Fisher) at 56°C. After 1 hour the solution was replaced with 100% Histoplast paraffin wax. After several changes of 100% wax, hearts were oriented and embedded into a mould pre-loaded with paraffin. The wax was then rapidly cooled on a bed of ice and stored at 4°C. A microtome was used to cut 10µm-thick sections through the heart. Before staining, sections were first deparaffinised in HistoClear solution (Fisher), followed by rehydration through a decreasing concentration of ethanol. For Masson's Trichrome Staining, the Masson's Trichrome kit (Abcam) was used as per the manufacturer's instructions. Briefly, slides were immersed in Bouin's solution for 15 minutes at 56°C. Sections were then stained by serial immersion in the following solutions: Weigert's Iron Haematoxylin solution (5 minutes); Biebrich Scraletacid Fuchsin (5 minutes); Aniline Blue (5 minutes); Phosphotungstic/Phosphomolybdic acid solution (5 minutes) and finally 1% Acetic acid solution (2 minutes). Samples were then rinsed in distilled water and dehydrated. Finally, sections were cleared in Xylene and mounted using DPX mounting media.

Immunofluorescence staining

Embryos were harvested at the required embryonic stage, placed in ice-cold PBS (Sigma) and the heart micro-dissected. Similarly, hearts from neonates were removed and washed in ice-cold PBS. Both embryonic and neonatal hearts were fixed for 6 hours in 2% paraformaldehyde (PFA; Santa Cruz Biotechnology) at 4°C and equilibrated in 30% sucrose overnight at 4°C. Hearts were then placed in 50:50 30% sucrose/PBS: Tissue-Tek OCT (VWR) for 30 minutes at room temperature (RT) and embedded in OCT. 10 µm-thick cryosections through the heart were cut. Before use, slides were left to dry for 10 minutes at RT and then washed in PBS for 5 minutes to remove the OCT. Samples were permeabilised with 0.5% (sections) or 0.1% (explants) Triton X-100 in PBS (PBTr) for 10 minutes at RT and subsequently rinsed twice in PBS. Samples were blocked in 10% Serum, 1% bovine serum albumin (BSA, Merck), 0.1% PBTr for 1 hour at RT prior to incubation with the primary antibodies overnight at 4°C. The following day, slides were washed three times for at least 5 min in 0.1% PBTr. Samples were incubated with Alexa Fluor®-conjugated secondary antibodies (1:200 dilution; Invitrogen) and 4',6-Diamidino-2-Phenylindole, Dihydrochloride (DAPI; 0.1 µg/ml, Invitrogen), for 1 hour at RT, protected from light. After final washes, slides were mounted in 50% glycerol in PBS. For wholemount staining, samples were washed in 0.3% PBTr and blocked in 1% BSA (Merck), 0.3% PBTr for at least 2 h. The samples were then incubated with primary antibodies in the blocking solution overnight at 4°C. On the second day, the samples were washed at least five times in 0.3% PBTr and then incubated with secondary antibodies and DAPI (Invitrogen) diluted in PBS overnight at 4°C. The samples were then washed with PBS at least five times the next day and mounted in 50% glycerol in PBS. The following primary antibodies were used: Dylight 549 Mab (Hypoxypore-1 Inc, 1:200) antibody to detect hypoxypore labelling, HIF-1α (1:100, Novus Biologicals), HIF-2α (1:100, R&D Systems), endomucin (1:50, Santa Cruz Biotech), podoplanin (1:200, Fitzgerald), WT1 (1:100, Abcam), Alexa Fluor™ 488 Phalloidin (1:250, ThermoFischer), actin alpha-smooth muscle Cy3 (1:100, Sigma), Anti-α-Actinin (Sarcomeric) (1:500, Sigma-Aldrich), CD31 (1:100, Abcam), smooth muscle Myosin heavy chain 11 (SM-MHC, 1:100, Abcam), SM22 alpha (1:100, Abcam), Wheat-germ agglutinin (WGA, 1:100, ThermoFisher), prollyl hydroxylase domain-2 (PHD2, 1:100, Novus Biologicals). Endomucin staining of coronary vessels was quantified using AngioTool™.

RNA isolation and qRT-PCR analysis

Total RNA was isolated from frozen ventricles and epicardial explants with Trizol (Invitrogen) using a Teflon homogeniser followed by aspiration with a sterile 25-gauge needle and syringe (BD sciences). RNA was transcribed into cDNA utilizing random primers (Promega) with Superscript

Reverse Transcriptase III (Life Technologies). Real-time quantitative PCR was performed on a ViiA™ 7 Real-Time PCR System (Applied Biosystems), using SYBR Green mix (Invitrogen). Gene expression was evaluated as DeltaCt relative to control (*Atp5b*, *Sdha* and *B2m*).

Heart dissociation and FACS sorting

Hearts from P1 and P7 mice were harvested and minced into a single cell suspension of FACS sorted live cells. Briefly, for each sample, 3 hearts collected from the same litter were pooled together and finely minced with a scalpel. Tissue was then digested by gentle agitation (180 rpm shaker) in Collagenase II (Gibco,) using a solution of 500 units/ml in HBSS at 37 °C for 45 minutes (P7 hearts) or 20 minutes (P1 hearts). Cell solutions were then passed through a 70 µm filter, washed and incubated in Red Blood Cell lysis buffer (Cell Signaling Technology) for 10 minutes at RT to remove red blood cells. Finally, isolated single cardiac cells were centrifuged, resuspended in 2% FBS in PBS, passed through the filtering cap of the FACS tubes and incubated with 1% 7-AAD viability stain (Invitrogen) for 10 minutes. Approximately 1×10^5 live cells per sample were sorted using the BD FACSAria Fusion Sorter.

Single cell RNA-Sequencing

Hearts were isolated from intact P1 and P7 mice. FACS sorted cardiac cells viability and concentration were assessed by using an automated cell counter and 1.5×10^4 cells per sample were loaded onto the 10X Chromium system (10X Genomics) to obtain a target cell recovery of ~6000 cells. Single cell RNA-seq libraries were generated using Single Cell 3 Prime Reagent Kits v1.3 (10x Genomics) according to the manufacturer's protocol. Sequencing was performed on an Illumina NovaSeq 6000 System operated by the Oxford Genomics Centre at the Wellcome Centre for Human Genetics. Raw sequence reads were aligned against the mouse mm10/GRCm38 reference transcriptome using the Cell Ranger 3.1.0 pipeline (10x Genomics) and processed further using the scRNA-seq analysis R package Seurat (v.3.2.3). Initial filtering removed cells expressing less than 200 genes and genes that were expressed in less than 3 cells. To exclude low quality cells and doublets we filtered out cells with a very high mitochondrial genes content and cells that expressed more than 8000 genes. Based on these criteria, 19211 genes across 5308 and 8734 cells for P1 and P7 samples respectively remained for downstream analysis. Data from the two samples were combined and scaled by regressing out the nUMIs, percentage of mitochondrial gene expression and cell cycle. To correct for batch effect, samples were integrated using the Harmony package (v.1.0). Uniform manifold approximation and projection (UMAP) was performed on the scRNAseq harmonised cell embeddings and unbiased clustering was obtained using the FindCluster function of the Seurat pipeline. Cluster cell types were annotated using a combination of differentially expressed markers, identified using the Seurat FindAllMarkers function and the expression of selected canonical markers for specific cell types. The Epicardial cell cluster was then identified and subset into a new Seurat object with raw reads counts. The standard Seurat pipeline described above was performed and the Model-based Analysis of Single-cell Transcriptomics test (MAST v. 1.14.0) was used to analyse the differential gene expression between the P1 and P7 groups in the epicardial cells. Finally, differentially expressed genes were ranked based on both the fold change and p value ($\text{avg_logFC} * -\log_{10} \text{ of the } p\text{-val}$) and the ranked list was submitted to the Metascape platform (www.metascape.org) to identify enriched pathways in the P1 or P7 group, respectively.

Cardiac cine-MRI

Cardiac cine-MRI was performed at 7T using a Varian DDR system. Briefly, mice were anaesthetised with 2% isoflurane in O₂ and positioned supine in a custom animal handling system with homeothermic control. Prospectively gated proton cardiac images were acquired with a partial Fourier accelerated spoiled gradient echo CINE sequence (TR 5.9 ms, TE 2.2 ms, 30 kHz bandwidth, 25° FA, approximately 20-30 frames gated to the R wave with a 4 ms post-label delay; 20% partial acquisition; 4 averages) with a 72 mm volume transmit/4 channel surface receive coil

(Rapid Biomedical GmbH) in order to acquire two and four chamber long-axis views and a short axis stack for functional quantification (128×128 matrix; 25.6 mm² FOV; 0.7 mm slice thickness, 0.2 mm resolution in-plane). Non-acquired partial Fourier data was reconstructed via the method of projection onto convex sets prior to a simple, cartesian, DFT. Blinded image analysis was performed in ImageJ as described previously. Mice with either excessive or minimal infarct sizes were excluded from further analyses by MRI scanning at day 1 post-MI.

Wound healing/scratch assay

2×10^4 MEC.1 cells were seeded on a 2 well Culture insert (Ibidi, Germany). The day after, the insert was removed and 50 µM of BAY 85-3934 (Molidustat, Selleck) or vehicle (DMSO) was added to the culture. Cells were imaged every 2 hours up to 6 hours. The cell-free space area was determined by using the ImageJ software (NIH, Rockville, USA).

Image Analysis

For quantification of immunofluorescence of cryosections, images were captured on an Olympus FluoView 3000 confocal microscope and analysed with Fiji (NIH) and AngioTool⁷. To quantify nuclear fluorescence, a macro was written to identify all DAPI-stained nuclei and to analyse nuclear fluorescence based on the DAPI mask. To localise WT1+ cells in the epicardium and myocardium, cells were manually counted. To quantify the cell size, three independent samples per group with six different fields and positions, two from left and right ventricles, and septum were captured at 40× magnification. ImageJ (National Institutes of Health) was used to quantify the size of each cell.

Statistical Analysis

All data are presented as mean ± standard error of the mean (SEM) or as median, inter-quartile range (IQR) and upper and lower limits. Statistical analysis was performed on GraphPad Prism 8 software. The statistical significance between two groups was determined using an unpaired two-tailed Student's t-test, these included an F-test to confirm the two groups had equal variances. Among three or more groups, one-way analysis of variance (ANOVA) followed up by Dunnett's or Tukey's multiple comparison tests were used. A value of $p \leq 0.05$ was considered statistically significant.

Data availability

The scRNA-Seq dataset analysed herein has been uploaded into the Gene Expression Omnibus (GEO) repository, accession number GSE305621.

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We are very grateful to Prof. Sir Peter J Ratcliffe for technical advice and highly informative scientific discussions throughout the project. We would like to thank Biomedical Services Unit for animal husbandry and Micron Oxford Advanced Bioimaging Unit (supported by Wellcome Strategic Awards 091911/B/10/Z and 107457/Z/15/Z) for access to and training in the use of confocal microscopy.

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Reviewer #1 (Public review):

Summary:

The manuscript by Gamen et al. analyzed the functional role of HIF signaling in the epicardium providing evidence that stabilization of the hypoxia signaling pathway might contribute to neonatal heart regeneration. By generating different conditionally mouse mutants and performing pharmacological interventions, the authors demonstrate that stabilizing HIF signaling enhances cardiac regeneration after MI in P7 neonatal hearts.

Strengths:

The study presents convincing genetic and pharmacological approaches on the role of hypoxia signaling enhance the regenerative potential of the epicardium

Weaknesses:

The major weakness remains the lack of convincing evidence demonstrating the role of hypoxia signaling in EMT modulation in the epicardial cells. The authors claimed that EMT assays adopted in this study are based on similar previous studies. Surprisingly, two of the references provided correspond to their own research group (PMID: 17108969, PMID: 19235142), limiting the credit for such claims, and the other two (PMID: 27023710, PMID: 12297106) assessment of cell migration but not EMT is reported. Thus, EMT remains to be convincingly demonstrated.

<https://doi.org/10.7554/eLife.107419.2.sa3>

Reviewer #2 (Public review):

Summary:

In this study, Gamen et al. investigated the roles of hypoxia and HIF1a signaling in regulating epicardial function during cardiac development and neonatal heart regeneration. The authors identified hypoxic regions in the epicardium during development and demonstrated that genetic and pharmacological stabilization of HIF1a during neonatal heart injury prolonged epicardial activation, preserved myocardium, enhanced infarct resolution, and maintained cardiac function beyond the normal postnatal regenerative window.

Strengths:

HIF1a signaling was manipulated in an epicardium-specific manner using appropriate genetic tools.

Weaknesses:

Some conclusions still need clarification.

Comments on revisions:

(1) The authors' comment on the partial overlap of HP1 and HIF1a IF signals (HIF1a is highly unstable ... broader regions of hypoxia) is reasonable and would help readers interpret the data if included in the text describing Fig. 1.

(2) The conclusion regarding WT1+ cells in Fig. 2a and b remains unclear. Both panels display larger and smaller magenta cells, and when all are taken into account, the overall number does not appear substantially different. Additional clarification is needed on how the quantification was performed.

(3) Regarding Figure 6-figure supplement 1c, it seems difficult to conclude the endothelial identity of WT1+ cells based on EMCN staining, as the markers do not overlap. The authors note that WT1 is upregulated in endothelial cells, but this has been reported in the context of injury, which differs from the context of the present study involving Molidustat.

<https://doi.org/10.7554/eLife.107419.2.sa2>

Reviewer #3 (Public review):

Summary:

The author's research here was to understand the role of hypoxia and hypoxia-induced transcription factors Hif-1a in the epicardium. The authors noted that hypoxia was prevalent in the embryonic heart and this persisted into neonatal stages until post natal day 7 (P7). Hypoxic regions in the heart were noted in the outer layer of the heart and expression of Hif-1a coincided with the epicardial gene WT1. It has been documented that at P7, the mouse heart cannot regenerate after myocardial infarction and the authors speculated that the change in epicardial hypoxic conditions could play a role in regeneration. The authors then used genetic and pharmacological tools to increase the activity of Hif genes in the heart and noted that there was a significant improvement in cardiac function when Hif-1a was active in the epicardium. The authors speculated that the presence of Hif-1a improved cell survival.

Strengths:

A focus on hypoxia and its effects on the epicardium in development and after myocardial infarction. This study outlines a potential to extend the regenerative time window in neonatal

mammalian hearts.

Weaknesses:

While the observations of improved cardiac function is clear, the exact mechanism of how increased Hif-1a activity causes these effects is not completely revealed. The authors mention improved myocardium survival, but do not include studies to demonstrate this.

There is an indication that fibrosis is decreased in hearts where Hif activity is prolonged, but there are no studies to link hypoxia and fibrosis.

Comments on revisions:

In the manuscript revision, the authors address my comments. They outline differences between genetic disruption of Phd2 and chemical inactivation could be due to dosing and drug half-life of Molidustat. The other comments are addressed by explaining that they have analyzed enough heart sections and hearts to come to their conclusions. The authors also state they cannot generate more numbers for this study, therefore I accept their conclusions as stated.

<https://doi.org/10.7554/eLife.107419.2.sa1>

Author response:

The following is the authors' response to the original reviews

eLife Assessment

This valuable study investigates the role of HIF1a signalling in epicardial activation and neonatal heart regeneration in mice. Through a combination of genetic and pharmacological approaches, the authors show that stabilization of HIF1a enhances epicardial activation and extends the regenerative capacity of the heart beyond the typical neonatal window following myocardial infarction (MI). However, several aspects of the study remain incomplete and would benefit from further clarification and additional experimental support to solidify the conclusions.

We reveal herein prolonged epicardial activation following myocardial infarction (MI) beyond post-natal days 1-7 (P1-P7) by genetic or pharmacological stabilisation of HIF-signalling. This extends the so-called “regenerative window” during an adult-like response to injury, leading to enhanced survived myocardium and functional improvement of the heart, even against a backdrop of persistent, albeit reduced, fibrosis. The epicardium is known to enhance cardiomyocyte proliferation and myocardial growth during heart development via trophic growth factor (for example, IGF-1, FGF, VEGF, TGF β and BMP) signalling (reviewed in PMID:29592950) and epicardium-derived cell-conditioned medium reduces infarct size and improves heart function (PMID: 21505261). Further experiments, outside of the scope of the current study, are required to determine whether activated neonatal epicardium elicits similar paracrine support to sustain the myocardium and heart function after injury beyond P7 into adulthood.

Public Reviews:

Reviewer #1 (Public review):

Summary:

The manuscript by Gamen et al. analyzed the functional role of HIF signaling in the epicardium, providing evidence that stabilization of the hypoxia signaling pathway might contribute to neonatal heart regeneration. By generating different conditionally mouse mutants and performing pharmacological interventions, the authors demonstrate that stabilizing HIF signaling enhances cardiac regeneration after MI in P7 neonatal hearts.

Strengths:

The study presents convincing genetic and pharmacological approaches to the role of hypoxia signaling in enhancing the regenerative potential of the epicardium.

Weaknesses:

The major weakness is the lack of convincing evidence demonstrating the role of hypoxia signaling in EMT modulation in epicardial cells. Additionally, novel experimental approaches should be performed to allow for the translation of these findings to the clinical arena.

We respectfully disagree that we have not convincingly demonstrated a role for HIF-signalling in promoting epicardial EMT. We adopt epicardial explant assays utilising a well characterised ex vivo protocol previously described for studying EMT in embryonic, neonatal and adult epicardium (PMID: 27023710, PMID: 12297106; PMID: 17108969, PMID: 19235142). These assays demonstrate in WT1^{CreERT2};Phd2^{fl/fl} explants enhanced cobblestone to spindle-like change in cell morphology, increased cell migration, appearance of stress fibres and an up-regulation of the mesenchymal marker alpha-smooth muscle actin (α SMA); all parameters associated with EMT. In addition, our in vivo analyses of Wt1^{CreERT2};Phd2^{fl/fl} hearts, in response to neonatal injury, reveal elevated numbers of WT1⁺ epicardial cells within the sub-epicardial region and underlying myocardium as is associated with active EMT and subsequent migration from the epicardium.

Reviewer #2 (Public review):

Summary:

In this study, Gamen et al. investigated the roles of hypoxia and HIF1a signaling in regulating epicardial function during cardiac development and neonatal heart regeneration. They found that WT1⁺ epicardial cells become hypoxic and begin expressing HIF1a from mid-gestation onward. During development, epicardial HIF1a signaling regulates WT1 expression and promotes coronary vasculature formation. In the postnatal heart, genetic and pharmacological upregulation of HIF1a sustained epicardial activation and improved regenerative outcomes.

Strengths:

HIF1a signaling was manipulated in an epicardium-specific manner using appropriate genetic tools.

Weaknesses:

There appears to be a discrepancy between some of the conclusions and the provided histological data. Additionally, the study does not offer mechanistic insight into the functional recovery observed.

We respectfully disagree with the comment that our histological data does not support our conclusions and expand on this in the response to specific reviewer comments. We agree that further mechanistic experiments outside of the scope of the current study are required to

identify precisely how activated neonatal epicardium results in increased healthy myocardium after injury beyond post-natal day 7 (P7).

Reviewer #3 (Public review):

Summary:

The authors' research here was to understand the role of hypoxia and hypoxia-induced transcription factor Hif-1a in the epicardium. The authors noted that hypoxia was prevalent in the embryonic heart, and this persisted into neonatal stages until postnatal day 7 (P7). Hypoxic regions in the heart were noted in the outer layer of the heart, and expression of Hif-1a coincided with the epicardial gene WT1. It has been documented that at P7, the mouse heart cannot regenerate after myocardial infarction, and the authors speculated that the change in epicardial hypoxic conditions could play a role in regeneration. The authors then used genetic and pharmacological tools to increase the activity of Hif genes in the heart and noted that there was a significant improvement in cardiac function when Hif-1a was active in the epicardium. The authors speculated that the presence of Hif-1a improved cell survival.

Strengths:

A focus on hypoxia and its effects on the epicardium in development and after myocardial infarction. This study outlines the potential to extend the regenerative time window in neonatal mammalian hearts.

We thank the reviewer for this positive endorsement and recognition of the importance of mechanistic insight into how to extend the window of neonatal heart regeneration.

Weaknesses:

While the observations of improved cardiac function are clear, the exact mechanism of how increased Hif-1a activity causes these effects is not completely revealed. The authors mention improved myocardium survival, but do not include studies to demonstrate this.

We report an increase in healthy myocardium arising from prolonged activation of the epicardium during the neonatal window and following injury at post-natal day 7 (P7). We speculate this recapitulates the role of the epicardium during heart development which is known to be a source of trophic growth factors that can enhance myocardial growth. Further experiments are required, out-of-scope of this study, to define a mechanistic link between HIF-signalling, epicardial activation and myocardial survival in the setting of prolonged neonatal heart regeneration.

There is an indication that fibrosis is decreased in hearts where Hif activity is prolonged, but there are no studies to link hypoxia and fibrosis.

We believe the decreased fibrosis is a natural consequence of the increase in survived myocardium arising from the activated epicardium. There is strong precedent here following injury at post-natal day 1 (P1) in which fibrosis is evident early-on but is resolved over time with growth of the myocardium in the regenerating heart (PMID: 23248315).

Recommendations for the authors:

Reviewing Editor Comments:

(1) Address issues related to image quality, colocalization, sample labeling, appropriate controls, and quantification - particularly in Figures 1, 2, 6, and Supplementary Figure 9.

Increase sample size as noted by reviewers.

The issues of co-localisation and sample labelling have been addressed under response to reviewers. We are unable to increase sample numbers but have clarified the number of regions per section and numbers of sections per heart analysed where appropriate.

(2) Clarify the effects of epicardial HIF1a activation on neovascularization.

We have removed reference in the abstract to an effect on neovascularisation.

(3) Extend assessments of epicardial hypoxia and HIF1a expression to earlier embryonic stages, when epicardial EMT is more active.

Our earliest timepoint of E12.5 marks the onset of epicardial EMT and E13.5 is the stage with the most significant mobilisation of epicardium-derived cells (EPDCs) into the sub-epicardial region and underlying myocardium (PMID: 32359445). In the same study, E11.5 lineage tracing of epicardial cells is restricted to outer layer of the heart; thus, our timepoints are representative in capturing both the onset and progression of in vivo EMT.

(4) Strengthen EMT assays and mechanistic modeling. Provide evidence from physiologically relevant models, as current 2D culture assays do not adequately support conclusions about EMT. Include additional EMT markers and quantification where appropriate.

We respectfully disagree that epicardial explants are not a valid assay for assessing EMT. As noted under responses to reviewers, such primary explants have been widely described elsewhere (PMID: 27023710, PMID: 12297106; PMID: 17108969, PMID: 19235142) and enable documentation of multiple parameters that are associated with active EMT, including an assessment of the extent of cell migration, cobblestone (epithelial) to spindle-like (mesenchymal) cell morphologies, stress fibre formation and expression of alpha-smooth muscle actin as a mesenchymal marker. We support our findings in explants by revealing reduced WT1⁺ epicardium-derived cells (EPDCs) in the sub-epicardial region and underlying myocardium of WT1^{CreERT2/+};Hif1a^{fl/fl} embryonic hearts (data in Figure 2) indicative of impaired epicardial EMT and migration of EPDCs and in vivo following neonatal MI with pharmacological inhibition of PHD2, where we observe the reciprocal phenotype of increased numbers of epicardium-derived cells emerging from the outer epicardial layer (data in Figure 6).

(5) Strengthen mechanistic insights into the role of epicardial cells in the functional recovery observed in MI hearts.

We agree that further experiments are required, out-of-scope of this study, to define a mechanistic link between HIF-signalling, epicardial activation and myocardial survival in the setting of prolonged neonatal heart regeneration.

Reviewer #1 (Recommendations for the authors):

The manuscript by Gamen et al. analyzed the functional role of HIF signaling in the epicardium, providing evidence that stabilization of the hypoxia signaling pathway might contribute to neonatal heart regeneration. By generating different conditionally mouse mutants and performing pharmacological interventions, the authors demonstrate that stabilizing HIF signaling enhances cardiac regeneration after MI in P7 neonatal hearts. The study is potentially interesting, but it presents several major caveats.

(1) One of the critical points reported in the early stages of this study is the early co-localization of Wt1, the hypoxic report (HP1), and HIF signaling pathways master regulators (i.e., HIF1a and HIF1b) during embryonic development. Figure 1 is meant to report such findings. However, unfortunately, I hardly see any co-localization at all in the Wt1+ epicardial cells for HP1, with some colocalization is seen for HIF1 and 2 alpha, although none of these data are quantified. Thus, it is hard to believe such co-localization.

We respectfully disagree with this comment. We highlight cells in Figure 1 that are co-stained for WT1+ and HP1. In addition, we identify HIF1-α and HIF2- α positive cells which either reside within the epicardium, as the outer cell layer, or within the underlying sub-epicardial region, respectfully.

(2) The authors claimed that they have analyzed the expression of the hypoxic report, as well as Wt1 and the HIF signaling pathways master regulators (i.e., HIF1a and HIF1b) in the AV groove, as compared to the apex, in embryonic heart ranging from E12.5 to E18.5 (Figure 1). Unfortunately, all images provided that are tagged as AV groove are rather misleading. They do not represent the AV groove but part of the right ventricular free wall. If the authors want to refer to the AV groove, AV cushions should be visible underneath.

We have removed specific reference to the AV groove and refer to the highlighted regions as the “Base” of the heart.

(3) The authors analyzed the hypoxic condition of the developing heart from E12.5 to E18.5. However, it remains unclear why the authors only explored the hypoxic conditions from E12.5 onwards, since epicardial EMT mainly occurs earlier than this time point, i.e., E10.5 onwards. Therefore, it would be needed to explore it already at this earlier time point.

We respectfully disagree with the reviewer and refer to the comment above regarding the fact that E12.5 marks the onset of epicardial EMT and E13.5 is the stage with the most significant mobilisation of epicardium-derived cells (EPDCs) into the sub-epicardial region and underlying myocardium (PMID: 32359445).

(4) The authors reported a conditional mouse model of HIF1alpha deletion by using the Wt1CreERT2 driver. Curiously, Wt1 is dependent on hypoxia signaling (i.e., HIF1a). Therefore, it is unclear whether there is a negative feedback loop between the deletion of Hif1alpha and the activation of the Cre driver might have functional consequences. Convincing evidence should be provided that such crosstalk does not interfere with Hif1alpha inactivation, and therefore, appropriate controls should be run in parallel.

We discount a negative feedback loop in this instance based on the fact we have utilised heterozygous mice for the WT1^{CreERT2/+} line and observe a consistent and reproducible phenotype for the developing hearts on a Wt1^{CreERT2/+};Hif1a^{fl/fl} background and following injury in Wt1^{CreERT2/+};Phd2^{fl/fl} mice. Collectively this indicates that the WT1-CreERT2 driver is active in the context of diminishing HIF-1α and Phd2, respectively. In addition, have carried out parallel experiments using epicardial explants derived from R26R-CreERT2;Phd2^{fl/fl} (Figure 3) to circumvent any potential confounding issues; the results of which are consistent with increased epicardial EMT in support of our overall hypothesis.

(5) On Figure 2a-f the authors reported that epicardial cells are diminished in Wt1CreERT2Hif1alpha mice as compared to controls. I am very sorry, but I do not see

any difference. Furthermore, it is unclear to me how the authors quantified such differences, i.e., what marker signal did they use and how it was performed (Figure 2c and d)?

We respectfully disagree with the reviewer and draw attention to the single channel panels of WT1+ staining in Figure 2, which show clear differences between numbers of epicardial cells in the mutant mice compared to controls (comparing magenta cells in panels a) versus b). Quantification was carried out for numbers of WT1+ cells residing within the PDPN-positive epicardium (and underlying PDPN-negative myocardium) across multiple images from multiple sections and multiple hearts.

(6) On Figure 2g, the authors reported differences in total vessel length. Are they referring to impaired microvasculature development? Or is this analysis also including major coronary vessels? What about the major coronary vessels and trees, is there any affection?

This analysis refers to the microvasculature and not the major coronary arteries or coronary trees.

(7) The authors reported that there might be some differences in EMT markers, but unfortunately, all of them are analyzed on 2D cultures, where no substrate for EMT is present, i.e., an underlying ECM bed. Thus, the authors cannot claim that EMT is altered. Additional experiments using either collagen substrate and/or Matrigel are required to fully demonstrate that EMT is impaired. Furthermore, quantitative analyses of such differences should be provided.

The 2D cultures are epicardial explants from mutant versus wild type hearts and represent a widely adopted previously published ex-vivo assay for investigating epicardial EMT across embryonic to adult stages (PMID: 27023710, PMID: 12297106; PMID: 17108969, PMID: 19235142); including an assessment of the extent of migration and cobblestone (epithelial) to spindle-like (mesenchymal) cell morphologies, stress fibre formation and expression of alpha-smooth muscle actin as a mesenchymal marker. We do not understand the comment regarding an “underlying ECM bed” as the cells exhibit EMT routinely on tissue culture plastic and will deposit their own ECM during the culture time course and in response to EMT/cell migration. In terms of quantification this was carried out for scratch assay experiments, as a proxy for EMT and emergent mesenchymal cell migration, as presented in Figure 3i, j with significant enhanced scratch closure and cell migration following Molidustat treatment.

(8) The description of data provided on Supplementary Figure 5 is spurious and should be removed. A note in the discussion might be sufficient.

We respectfully disagree. The ChIP-seq data, in what is now Figure 2- figure supplement 3, highlights a HIF-1 α binding site within the *Wt1* locus suggesting putative upstream regulation of WT1 by HIF-1 α . Thus this provides a potential explanation as to how HIF-1 α may activate the epicardium through up-regulation of *Wt1*/WT1.

*(9) On Figure 3, the authors further illustrate the change of EMT markers using ex vivo cardiac explants. They reported increased expression of *Snai2* that, although statistically significant, is most likely of no biological relevance (increase of only 20% at transcript level). What about *Snai1*, *Prrx1*, and other EMT promoters? Are they also induced? As previously stated, these 2D cultures do not provide supporting evidence that EMT is occurring, thus 3D gel assays should be performed in which Z-axis analyses will provide evidence on the different migratory behaviour of those cells.*

We respectfully suggest that a 20% change in *snai2* expression is biologically meaningful with respect to EMT. This in-turn is supported by associated cell migration, reduced ZO-1 expression, increased stress fibres and increased alpha-SMA as a mesenchymal marker; all properties associated with active EMT. Other suggested markers have not been validated as formally required for EMT, for example *Snai1* (PMID: 23097346). The migratory capacity of targeted versus epicardial cells was assessed by combined explant and scratch assay experiments.

(10) The description of single-cell analyses is very incomplete. Which mice were used for these analyses, wildtype control, or hypoxic mice? Please provide a clearer description of the samples used. Additionally, the entire rationale of these analyses is dubious. Doing single-cell analyses to analyze a couple or three markers in a very small cell population is rather ridiculous. qPCR might be far more appropriate and convincing, or a bulk RNAseq analysis of isolated epicardial cells.

The single-cell analyses represent an unbiased assessment of different pathways in epicardial cells (identified bioinformatically) between intact P1 and P7 stages in wild type (control) hearts, with a focus on hypoxia-related gene expression and HIF-dependent pathways. It was not designed to analyse a small number of genes, rather global differences in the hypoxic states between P1 and P7 hearts. Selected genes (*Vegfa*, *Pdk3*, *Egln 1* (*Phd2*)) were analysed to highlight the key differences in hypoxic signalling across the regenerative window. The fact the hearts were uninjured/intact is clarified in the text and legends for Figure 4 and now Figure 4-figure supplement 1.

(11) The analyses provided in Figure 5 are very interesting and their findings are very relevant. However, I would think that the complementary experimental approach should also be done, i.e, MI followed by activation with tamoxifen, since that situation would be more realistic in the clinical setting.

Tamoxifen causes respiratory failure in neonates with MI, so the two cannot be combined at the same time or soon after surgery. Moreover, tamoxifen takes significant time to take effect on targeted gene down-regulation which may negate sufficient activation of the epicardium following injury.

The experiments in Figure 5 were designed to demonstrate that prolonged heart regeneration could be elicited in a cell-specific (epicardial-specific) manner via a genetic approach. The pharmacological experiments in Figure 6 are complementary in this regard by demonstrating equivalent effects with drug (Molidustat) delivery to reduce PHD2 and stabilise HIF post-MI.

(12) In Figure 6, expression of Wt1 is highly prominent in P7 controls, mainly restricted to the epicardial lining while in the experimental setting, such Wt1 expression is broadly distributed on the subepicardial space, nicely demonstrating epicardial activation. However, it is very surprising to see such Wt1 expression in controls, something that is not expected, as compared to the data reported in Figure 4g. Could the authors please reconcile these findings?

Figure 6 represents the injury setting and Figure 4g the intact setting (as clarified above, in the text and revised figure legends). Hence in the latter WT1 expression is significantly reduced in the P7 heart, as anticipated. With injury at P7 we anticipate activation of WT1 in control hearts, albeit restricted to the epicardial layer (as occurs in adult hearts, PMID: 21505261). In contrast, following Molidustat-treatment of P7 hearts post-MI we observe extensive epicardial expansion into the sub-epicardial region and EPDC migration into the underlying myocardium (Figure 6b).

Reviewer #2 (Recommendations for the authors):

The role of hypoxia and HIF1a signaling in epicardial activation is an important topic, and the genetic approaches employed in this study are appropriate. However, several aspects of the study remain unclear and would benefit from further clarification or explanation by the authors:

(1) The authors detected hypoxic regions using an anti-pimonidazole fluorescence-conjugated monoclonal antibody (HP1). The data would become more compelling if negative and positive controls were provided.

We believe the HP1 staining is compelling in the images shown and is consistent with hypoxic regions of the developing heart. We reveal HP1 staining at cellular resolution with neighbouring cells positive and negative for the HP1 signal in the apex of the heart and within the epicardium and sub-epicardial regions at E12.5 (Figure 1a) and diminished/altered hypoxic/HP1 regional signal through subsequent developmental stages at E14.5-18.5 (Figure 1a-d).

(2) Many HIF1a-positive cells in the AV groove region do not appear to overlap with HP1 staining (Figure 1a). Providing a low-magnification image of HIF1a expression would be helpful to better assess the extent of overlap with HP1 staining

HIF-1 α is highly unstable and hence detection of HIF-1 α + cells will likely only sample of cells compared to HP1 which is a surrogate for broader regions of hypoxia.

(3) Although the authors conclude that epicardial HIF1a deletion results in a significant reduction of WT1⁺ cells in both the epicardium and myocardium (Figure 2a-d), the provided images are not sufficiently clear to fully support this interpretation. Providing additional evidence to support this conclusion would be helpful.

We respectfully disagree with the reviewer and draw attention to the single channel panels of WT1+ staining which show clear differences between numbers of epicardial cells in the mutant mice compared to controls (Figure 2a versus 2b; magenta WT1+ staining).

(4) Similar to the point raised above, the authors' conclusion regarding the increased expression of WT1 following Molidustat treatment does not appear to be fully supported by the provided images (Figure 6b-f). Immunofluorescence staining for WT1 does not clearly demonstrate epicardial expression in the remote zone of either the control or Molidustat-treated hearts. In addition, while an increase of WT1⁺ cells is observed in the infarct zone of the Molidustat-treated heart, it is somewhat unexpected that such expansion is not evident in the corresponding region of the control heart, given that epicardial cells typically expand near the infarct area. Clarification on these points would be helpful.

Figure 6b reveals WT1 expression in controls (upper panel set) that is reactivated proximal to the infarct region, given WT1 is not expressed in adult epicardium but restricted to the epicardial layer (as occurs in injured adult mouse hearts PMID: 21505261). This contrasts with what is observed in the Molidustat-treated P7 hearts post-MI, where we observe epicardial expansion and migration of WT1+ cells into the underlying myocardium (Figure 6b, lower panel set, infarct zone).

(5) The authors conclude that WT1⁺ cells in the myocardial tissue exhibit endothelial identity based on the colocalization of WT1 and EMCN signals (Supplementary Figure 9c). However, this interpretation is difficult to assess, as WT1 is a nuclear marker and EMCN is

a membrane protein, which makes precise colocalization challenging to confirm with confidence. Additional supporting evidence may be necessary to substantiate this conclusion.

WT1 is known to be up regulated in endothelial cells in response to injury as shown previously in several studies (for example, PMID: 25681586). Here we show clear co-localisation of nuclear WT1 and cytoplasmic Endomucin (EMCN) in what is now Figure 6-figure supplement 1c and would encourage the reviewer and readers to magnify the image by zooming-in on the relevant co-stained panel.

(6) The authors conclude that activation of epicardial HIF1a signaling has no effect on neovascularization in postnatal MI hearts (Figure 5c). However, the abstract states: "Finally, a combination of genetic and pharmacological stabilisation of HIF ... increased vascularisation, augmented infarct resolution and preserved function beyond the 7-day regenerative window" (Lines 38-41). Clarification regarding this apparent discrepancy would be appreciated.

The abstract has been altered to remove the statement of increased vascularisation.

(7) The study appears somewhat incomplete, as it lacks mechanistic insight into the functional recovery observed following epicardial Phd2 deletion and Molidustat treatment in postnatal MI hearts. Although the authors suggest a potential paracrine role of the epicardium in protecting cardiomyocytes from apoptosis, this hypothesis has not been experimentally addressed. Incorporating such analysis would help to reinforce the study's conclusions.

Further experiments are required, which are out-of-scope of this study, to define a mechanistic link between the genetic or pharmacological stabilisation of HIF-signalling, epicardial activation and myocardial survival in the setting of prolonged neonatal heart regeneration.

Other points:

(1) Providing single-channel images for Figures 1a-d and 6g would be helpful for clarity and interpretation.

We believe the combined channel views of co-staining for two markers on a background of DAPI staining to pin-point cell nuclei, are informative and support our conclusions.

(2) Have the authors considered using AngioTool to quantify the number of vessels in Figure 5b-c?

AngioTool™ was used to quantify the vessels, as we have used previously (PMID: 33462113) and this is now added to the methods and legend of Figure 2.

Reviewer #3 (Recommendations for the authors):

There are several areas where the manuscript can be improved, such that its conclusions can be solidified.

(1) The authors highlight a point where blocking Phd2 can enhance survival of cardiac tissue, but did not report on survival markers. They surmised that apoptosis could be decreased in Phd2 mutant or Molidustat treatment but did not show this. The authors should determine if apoptosis is decreased in the myocardium and epicardium.

We show evidence of increased levels of healthy myocardium in the genetic and pharmacological models of stabilised HIF-signalling. We exclude increased cardiac hypertrophy or increased cardiomyocyte proliferation as causative, so suggest as a reasonable alternative enhanced survival, albeit this need not necessarily be via an apoptotic pathway given the incidence of necrotic cell death during MI. We are unable to generate new surgeries and mutant/treated heart samples to analyse for apoptotic markers at this stage.

(2) There appears to be no difference in cardiomyocyte proliferation in Molidustat-treated animals, but the experiment was only performed on 2 to 3 animals. This is too small a sample size to conclude from these results. The authors should increase the sample size to make this assertion.

We respectfully disagree that we are unable to conclude no effect on cardiomyocyte proliferation. We analysed multiple heart regions per section, for EdU+/cTnT+ colocalised signals across several sections per heart, set against a consistency of effect on other parameters in hearts treated with Molidustat. We are unable to generate more P7 heart surgeries +/- Molidustat and +/- EdU at this stage.

(3) It is curious as to how, after myocardial infarction, the fibrotic scar tissue is decreased in the Phd2 deletion but not as profound in Molidustat-treated mice at d21. Can the authors speculate why the difference exists and how this decrease arises? For example, are there decreased pro-inflammatory signals in Phd2 deleted mice? Is there decreased collagen deposition and ECM gene expression? Do macrophage recruitment into the infarct zone differ between mutant/treated vs WT?

The representative images in Figure 6k reveal a trend towards reduced fibrosis with Molidustat treatment (Figure 6l), but across all hearts analysed this was not as significant as observed in the epicardial-specific deletion injured hearts (Figure 5g, h). This may be due to the relatively short half-life of Molidustat (approximately 4-10 hours, PMID: 32248614), the dosing regimen for the drug and/or the fact that it was not specifically delivered/targeted to the epicardium.

(4) The magnified images in Figure 1 do not match the boxes in the whole heart images. It is unclear what the white boxes signify.

The white boxes have been removed from Figure 1. The magnified image panels are from serial heart sections and this is now clarified in the Figure 1 legend.

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